Holding Bankers Liable:

Personal Guarantees and Risk-Taking in Security Underwriting*

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Abstract

Does additional liability for bank executives improve risk management? We address this question using uniquely detailed data on the underwriting activities of one of the largest Dutch banks in the early 20th century. Its executives faced additional liability as they personally guaranteed part of their banks' security underwriting. If the issuance was successful, they received the underwriting fee; if it failed, they had to purchase securities at a loss. Exploiting exogenous discontinuities in the size of these guarantees, we find that issuance quality was higher for larger guarantees. Investor subscription rates were higher, issuing firms had stronger balance sheets, and subsequent security returns were higher. Results indicate that additional liability directly tied to executives' decisions can help reduce bank risk-taking.

Keywords: Limited liability, Executive compensation, Bank risk-taking, Financial history

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Recent banking crises have been linked to excessive risk-taking by banks and to executive pay that encouraged it.[1] Bank executives (henceforth "bankers") decide how much risk their banks take. Because banks are highly levered and their balance sheets are opaque, bankers can shift much of this risk onto depositors, creditors, and taxpayers. Their pay gives them an incentive to do so: bonuses and stock-based compensation reward the upside of risk-taking but expose bankers to little of the downside (Bebchuk and Spamann, 2010).

Abolishing bonuses and stock-based pay would curb this incentive, but it would also remove a reward for effort. Proposed reforms therefore seek to add personal liability to performance pay, so that bankers also bear part of the downside. Academic proposals tie bankers' pay or wealth to the bank's default risk, its distress, or its failure (Bolton et al., 2015; Acharya et al., 2016; Goodhart and Lastra, 2020). The U.K. and the EU have adopted rules that allow banks to claw back bonuses, and the United States has proposed similar rules.[2]

These reforms, adopted or proposed, rest on the premise that bankers take less risk when they bear more of the downside of their decisions. This is not obvious. Bankers' careers already depend on the bank's survival and on their reputation, so they may avoid excessive risk anyway (Amihud and Lev, 1981; Fama, 1980). Bankers may also misjudge the downside: they may be overly optimistic, or neglect tail risk altogether (Cheng et al., 2014; Gennaioli and Shleifer, 2018).

There is also a more specific reason why such reforms may fall short: the way liability is implemented. Most proposals tie it to the outcomes of the bank as a whole, which reflect the decisions of many people and which no single banker controls. Incentives work best when pay depends on outcomes that the agent's own actions strongly affect (Holmström, 1979, 1982). Clawback provisions do target individual bankers, but only after the fact. They require proving that a banker was responsible for a loss, which is hard in practice (Financial Stability Board, 2021). Liability is therefore more likely to change behavior if it is tied in advance to the decisions bankers actually make.

In this paper, we test this premise directly. We ask whether bankers take less risk when they are personally liable for the downside of their own decisions.[3] The ideal experiment

[1] On the 2008 crisis, see Financial Crisis Inquiry Commission (2011, pp. xviii–xix) and Group of Thirty (2009, p. 13). At Silicon Valley Bank, which failed in 2023, executive pay was tied to short-term earnings and included no risk metrics (Board of Governors of the Federal Reserve System, 2023).

[2] See European Parliament and Council of the European Union (2013) and Prudential Regulation Authority and Financial Conduct Authority (2015) for the EU and the U.K., and U.S. Government Accountability Office (2025) for the United States.

[3] Liability may also change who becomes a banker, because different pay schemes attract people with different abilities and attitudes toward risk (Dohmen and Falk, 2011; Bandiera et al., 2015). We compare the same bankers under different stakes, so our estimates capture only the change in their behavior.

would have two features. First, liability would be tied to individual decisions, each observed together with its outcome. Second, the size of that liability would vary for reasons outside the bankers' control.

Modern banks rarely offer either feature. Our setting comes close to this ideal: the security underwriting of the Nederlandsche Handel-Maatschappij (NHM), one of the largest Dutch banks, between 1919 and 1930. The NHM committed to buy any securities that investors did not take up, and its bankers personally guaranteed part of this commitment. On their part, the bankers faced the same payoff as the bank. If the securities sold, they earned its fee. If they did not, they had to buy their share of the securities that investors had declined. Their liability was thus tied to individual underwriting decisions, and we observe the outcome of each. No proof of fault was needed: the market's response to each issuance determined what the bankers gained or lost. The size of the guarantees followed a rule fixed in advance: they jumped whenever the amount the bank underwrote crossed one of a set of thresholds. We compare issuances below and above a threshold to estimate the causal effect of liability on what the bankers chose to underwrite.

Underwriting at the NHM worked much as firm-commitment underwriting does today. The NHM led or co-led the syndicates that underwrote each issuance. The underwriters paid the issuer a fixed price for the entire issuance and resold the securities to investors at a higher price. The difference was their fee. The NHM's bankers decided together, as its executive board, whether the bank would underwrite each issuance. On their guaranteed part, they bought unsold securities at the price the bank had paid the issuer.

The guarantees were a deliberate part of the NHM's pay policy, meant to raise the bankers' pay without encouraging excessive risk-taking. Like many banks today, the NHM kept fixed salaries low and paid its bankers mainly through a bonus set as a share of profits, to stimulate effort. The bank knew the danger in this arrangement. In 1879, a member of its supervisory board warned that bonuses could lead to "risky speculations, in which the bankers would share the profits, but not the losses."[4] Unlike a larger bonus, the guarantees made the bankers share in the losses as well as the profits, in an activity that was risky for the bank itself. Personal guarantees were a known practice among Dutch banks (Boissevain, 1891), and the NHM's chief executive made them standard at the bank after taking office in 1913 (De Graaf, 2012).

The sums at stake were large. A banker's average guarantee on a single issuance was about $f.$ 20,000, roughly a year's fixed salary.[5] The NHM led or co-led about 20 issuances a year, and the bankers provided guarantees on most of them. In our sample, 51% of

[4] NL-HaNA, NHM, 33: 21-11-1879, authors' translation.

[5] $f.$: Dutch guilder. The bankers' fixed salaries ranged from $f.$ 10,000 to $f.$ 35,000 a year (Appendix Figure B2).

issuances were not fully subscribed: too few investors were willing to buy at the price the underwriters asked. The bankers then had to sell their share of the unsold securities at a lower price or hold them in the hope that demand would recover.

Our key measure of risk is the subscription rate of each issuance, the share of the securities that investors bought. In underwriting, taking risk means committing the bank to issuances that might not sell. Because the guarantees made the bankers bear part of this risk, they had a reason to turn down issuances that were unlikely to sell, or to price issuances more conservatively. Either response raises the subscription rate. A single issuance could fail through bad luck. On average, however, the subscription rate reflects how likely an issuance was to fail when the bankers took it on.

Our empirical identification rests on the rule that set the guarantees. Under this rule, a banker's guarantee depended on the bank's underwriting amount in steps: it stayed constant over a range of amounts and rose, by $f.$ 6,000 on average, whenever the amount crossed one of a set of thresholds. The thresholds were round numbers. The archives contain no document that sets out the rule, so we infer it from the data. The rule stands out clearly: for more than half of all issuances, the guarantees equal exactly what it prescribes. If the guarantees had merely been rounded, far fewer would match the rule. Our estimates barely change when we infer the rule's guarantee levels in other ways.

Our analysis compares issuances of similar amounts below and above a threshold. These issuances differed mainly in how much the bankers had at stake. Because the bankers did not always follow the rule, we instrument each banker's actual guarantee with the guarantee the rule prescribes. Everything happens within one bank, so governance and regulation are held fixed, and banker fixed effects absorb changes in the board's composition. The bankers knew the rule and negotiated the bank's underwriting amount, but they could not change the rule itself. Any response to it, whether a safer issuance or a different amount, is therefore a consequence of the liability it imposed and part of the effect we measure.

Our data come from the bank's confidential executive minutes, now open to researchers. For each of the 239 issuances the NHM led or co-led, we observe the subscription rate, the bank's underwriting amount, and each banker's guarantee. We add the balance sheets of the issuing firms and monthly security prices, which we use to measure the quality of the issuers and the pricing and trading of their securities.

Larger guarantees raised the subscription rates of the issuances the bankers underwrote. At the average threshold, where the guarantee rose by $f.$ 6,000, the subscription rate rose by about 16%. When the NHM was the first-named lead and had most say over the issuance, the point estimate is about 27%. The effect is concave. Below a bank underwriting amount of $f.$ 2 million, an extra $f.$ 1,000 of guarantee raised the subscription

rate by about 8%; above it, by less than 2%. Where the bankers already had much at stake, a further increase added little, consistent with incentives that were already largely aligned. The bankers selected safer issuances: among firm issuances, those above a threshold came from firms with healthier balance sheets. They also priced them more conservatively. After the issuance, their securities traded more often, and their abnormal returns in the first month were 0.5 to 0.9 percentage points higher, against a sample mean of 3.1%. The guarantees were secret and could not be verified, so these outcomes cannot reflect a signal to investors. Bankers with more of their own money at stake thus took less risk.

Our results suggest that personal liability can make bankers take less risk, at least when it is tied in advance to individual decisions. Current policies do not take this form. In the United States, rules to defer and claw back incentive pay have been proposed since 2011 but never finalized, and bills to claw back pay from the executives of failed banks have stalled. U.K. and EU rules allow clawback only after the fact, once a banker's responsibility for a loss has been proven.[6] The NHM shows that liability can instead be fixed in advance for individual decisions, with no case to be made against a banker afterward. The concave effect also suggests that liability need not be large to matter: within the range we observe, the first guilders at stake did the most.

This paper contributes to three literatures, the first of which concerns the historical evidence on extended liability and bank risk. Esty (1998), Grossman (2001), and Mitchener and Richardson (2013) compare U.S. banks under different state liability laws, and Grossman and Imai (2013) compare British banks whose shareholders had different amounts of uncalled capital. Most of these studies find that more liability went with less risk.[7] Aldunate et al. (2021) compare state banks on either side of the border between single- and double-liability states during the Great Depression, and find that single liability raised the probability of bank distress. Koudijs et al. (2021), the study closest to ours, examine bank presidents whose personal liability depended on the marriage law in force when they married. Presidents whose wives' wealth was protected took on more leverage and lost more capital and deposits in the downturn of the 1870s. In each of these studies, liability was triggered only by the failure of the bank as a whole. Failure is an extreme outcome that depends on the decisions of many people and on the state of the economy, which no

[6]For the United States, see U.S. Government Accountability Office (2025), U.S. Senate (2023), and U.S. Senate (2026). For the U.K. and the EU, see European Parliament and Council of the European Union (2013), Prudential Regulation Authority and Financial Conduct Authority (2015), and Financial Stability Board (2024).

[7]Colvin (2018) finds that among Dutch banks lending to small and medium-sized firms in the 1920s, organizational form rather than shareholder liability predicted distress; the banks had chosen their own liability arrangements, and he notes that "the direction of causality is, however, open to debate" (p. 686). Diao et al. (2026) find that U.S. investment banks that converted from partnerships to limited liability from the 1970s onward underwrote riskier issuers, and note that these conversions "were not random and reflect conscious decisions by banks" (p. 6).

single banker controls. Relative to this literature, we study a form of liability that targets bankers' decisions more directly, tied to each decision before its outcome was known, and show that it reduced risk-taking.

Second, the paper relates to the literature on bank executive compensation. Every form of pay studied there ties a banker's exposure to the bank as a whole, and, as Edmans et al. (2017) stress, the endogeneity of pay contracts makes their causal effects hard to identify. Banks whose executives held larger equity stakes did worse in the 2008 financial crisis and failed more often (Fahlenbrach and Stulz, 2011; Berger et al., 2016). Inside debt aligns managers with creditors and is associated with lower risk (Sundaram and Yermack, 2007; Edmans and Liu, 2011; Wei and Yermack, 2011; Van Bekkum, 2016). Studies of post-crisis reforms, such as bonus caps and deferral, rely on a few large banks, and the reforms coincided with other regulatory changes (Kleymenova and Tuna, 2021; Colonnello et al., 2023; De Angelis et al., 2026; Gelos et al., 2026). Several of the authors describe their evidence as mixed or inconclusive. These reforms also put only variable pay at risk, not bankers' own wealth. Across firms more broadly, those that adopt clawbacks report more reliably and take less risk, but the decision to adopt them is endogenous (Chan et al., 2012; deHaan et al., 2013; Babenko et al., 2023). Our bankers' liability put their own wealth at stake, attached to individual decisions, and followed a fixed rule set in advance. As a result, we can estimate its causal effect.

Finally, the paper relates to work on managers' own stakes in other financial intermediaries. Mutual fund managers with larger stakes take less risk (Ma and Tang, 2019), and buyouts are less risky where private equity managers' stakes are large relative to their wealth (Bienz et al., 2023), though U.S. private equity shows no such relation (Robinson and Sensoy, 2013). In these settings, managers choose their stakes endogenously, and they may put their own money where they expect the best returns (Gupta and Sachdeva, 2025). Our bankers' stakes followed a fixed rule that they could not change.

The paper proceeds as follows. Section 1 describes the historical setting and the guarantees. Section 2 describes the data and how we infer the rule. Section 3 sets out the empirical approach, and Section 4 presents the results, including tests that validate the rule. Section 5 concludes.

Related Literature

This paper makes several contributions to the literature. First of all, we contribute to the literature on the effect of additional liability on bank risk-taking. Esty (1998), Grossman (2001), Mitchener and Richardson (2013), Calomiris and Carlson (2016) and Aldunate et al. (2021) find that additional liability for US banks before the Great Depression reduced risk-taking. For the UK, Grossman and Imai (2013) find that contingent liability in the form of uncalled capital led to less bank risk-taking, while Turner (2014) shows that

banks with unlimited liability were less likely to fail. Colvin (2018) examines the choice of liability arrangement in Dutch banks catering to small and medium sized businesses in the 1920s and finds that this mattered little for bank failure. Most closely related is Koudijs et al. (2021) who focus on the personal liability of bank executives. Bankers in New England in the nineteenth century that married under a regime that protected their wife's wealth were more likely to take on more risk and consequently lost more capital and deposits during the Long Depression (1873-1878). In contemporaneous work, Diao, Golubov and Pandes (2026) document that U.S. investment banks that chose to convert from partnerships to limited liability from the 1970s onward subsequently underwrote securities of riskier issuers. We contribute to this literature by analyzing a form of bankers' liability directly tied to individual decision making. Bankers might not be able to influence all bank risk-management directly. For instance, they might only have indirect control over deposit taking and the performance of the bank's entire loan portfolio, which are important determinants of a bank's overall profitability. In contrast, security underwriting is directly under their control. Our setting allows us to directly link additional liability to actual decision making and tests whether targeted additional liability is effective.

Second, our paper is related to literature on executive compensation in banking. Modern-day bankers' "skin-in-the-game" is in the form of equity compensation, which predominantly affects the upside of risk-taking due to the high leverage of banks (Bolton, Mehran and Shapiro, 2015). Empirically, higher equity stakes for bankers appear to increase bank risk (Fahlenbrach and Stulz, 2011; Berger, Imbierowicz and Rauch, 2016). At the same time, inside debt, e.g. future pensions, appears to reduce risk-taking for both firms (Wei and Yermack, 2011) and banks (Van Bekkum, 2016). We contribute to this literature by looking at the direct effects on risk taking of exposing bankers to downside risk.

Finally, our paper fits into the broader literature of managerial incentives and executive compensation. Edmans, Gabaix and Jenter (2017) provide an overview of theoretical and empirical work on executive compensation and argue that the causal effect of executive pay is difficult to identify empirically, due to the endogeneity of compensation contracts. We contribute to this literature by exploiting a predetermined step function in compensation contracts that bankers followed when underwriting security issuances.

The rest of the paper is structured as follows. Section 1 discusses the relevant historical background. Section 2 describes our newly constructed dataset. Section 3 details our empirical approach, while Section 4 presents our empirical results. Section 5 concludes.

1 Historical background

1.1 General background

During the late 19th and early 20th century, the financial sector in the Netherlands consolidated and large banks adopted a universal banking model. The "Big Five" banks, including the NHM, dominated the banking industry (De Jong and Roell, 2005), even though many specialized smaller banks still co-existed.[8]

The Netherlands remained neutral during the First World War and domestic businesses benefited substantially from exporting to foreign markets, where supply chains were disrupted (Van der Werf, 2014). Firms required ever more capital and the universal banks were quick to provide the funds by either extending bilateral loans or facilitating public issuance of securities. As in neighboring countries, issuing securities via underwriters was a standard practice for firms to attract large amounts of funding (Boissevain, 1891). A wide range of securities from private firms and public entities, both domestic and foreign, were issued and traded on the Amsterdam Stock Exchange. Different types of financial institutions were active in the security issuance business: brokers, private banking houses, general commercial banks and universal banks. The security market was free from any financial regulation or government intervention (Brouwer, 1969; Geljon, 2005).

After the First World War, the securities markets experienced a short boom in 1919-1920 before collapsing into a global post-war slump (Colvin, De Jong and Fliers 2015). Many banks failed due to the unprecedented credit boom from 1915-1920 and the consequent debt-deflationary period in the early 1920s, as their debtors were unable to repay their debts (Jonker, 1991; Colvin, De Jong and Fliers 2015; Jonker and Van Zanden, 1995). The resulting banking crisis between 1921 and 1924 led to the write-off of nearly 10% of the Big Five banks' assets (De Jong and Roell, 2005).[9] The crisis affected especially younger, smaller, less diversified and more leveraged banks, but even one of the Big Five

[8]The other large banks are Amsterdamsche Bank, Incasso-Bank, Rotterdamsche Bankvereeniging, and Twentsche Bank. Twentsche Bank merged with NHM in 1964 to become Algemene Bank Nederland, while Amsterdamsche Bank and Rotterdamsche Bankvereeniging merged as AMRO in the same year. These two merged banks united in 1991 to form ABN AMRO. For an overview of the Dutch banking system in the early twentieth century, see Colvin, De Jong and Fliers (2015).

[9]The exact start and end dates of the 1920s banking crisis are subject to debate. Some use only 1921 and 1922 as crisis years (Van Zanden, 1997; Bernanke and James, 1991). Gelderblom et al. (2023) mentions 1922 as starting year, while De Jong and Röell (2005) and Jonker (1991) see 1920 as the starting point, even though Jonker (1991) identifies 1922-1924 as the height of the crisis. Colvin (2018) defines 1920-1926 as crisis years. The bailout of one of the Big Five banks occurs in 1924 (Colvin, 2014) and Colvin, de Jong and Fliers (2015) show that recovery began in 1924 for stock prices of banks that did not experience distress, and in 1926 for distressed banks. We define the banking crisis to span the years 1921-1924. This decision is based on the number of security issuances during those years: in October 1920 the number of issuances starts declining (Van der Werf, 2014) and only in 1925 recovery sets in.

universal banks, Rotterdamsche Bankvereeniging, was on the brink of collapse and had to be bailed out.[10]

The economy and the stock market recovered in 1925, and a stock market boom followed. Industrial firms expanded their foreign operations, while Dutch banks increasingly underwrote foreign securities for both public and private issuers (Van der Werf, 2014). This upward trend persisted until 1930, when the Netherlands began to feel the impact of the Great Depression.

1.2 The Nederlandsche Handel-Maatschappij and Underwriting in the Dutch 1920s

The bank that we focus on in our analysis is the 'Nederlandsche Handel-Maatschappij' (Dutch Trading Society, NHM, the bank). The bank was one of the largest and most prestigious banks in the Netherlands in the early twentieth century. The bank was founded in 1824 as a limited liability colonial bank, engaging in financial and trading activities in the Dutch East Indies. Since its establishment the bank was renowned for its solid reputation (Wijtvliet, 1993).[11] Since the 1880s, the NHM shifted from a trading business towards universal banking activities, including securities underwriting.

The underwriting business was considered risky, and the bank's change of course to engage in investment banking in the 1880s was subject to intense discussion between the bankers and the supervisory board (De Graaf, 2012). In 1884, the bank adjusted its articles of association to allow for securities trading on its own account, which was a necessary legal requirement for the bank to be active in security underwriting. The bank created a provision balance sheet item as a buffer for potential losses from underwriting in 1888 and used this reserve already in 1889 (De Graaf, 2012).

We study the security underwriting activities of the NHM from 1919 until 1930. The 1920s were a particularly booming period for security issuance in the Netherlands. Universal banks were especially active, partly due to their large enough size to absorb potential losses from underwriting (Van Der Werf, 1990). The NHM was one of the largest players in the underwriting business. Before the First World War, the bank was responsible for more than half of the total syndicates and security issuances in the Netherlands (Kymmell, 1998). Three other large universal banks played an important role in the securities

[10] See Colvin (2014) for a description of the crisis and the near-collapse of the Rotterdamsche Bankvereeniging.

[11] The bank fulfilled a unique position in the Dutch society, as it had a special relationship with the government and the royal family and was not only a trading company, but also a commercial and investment bank. The colonial activities were very profitable, especially in the nineteenth century, and remained a substantial share of the bank's business in the first half of the twentieth century. See De Graaf (2012) for an elaborate overview of the bank's colonial business.

markets in this period as well (Amsterdamsche Bank, Rotterdamsche Bankvereeniging and Twentsche Bank). Many other smaller commercial banks and prestigious investment banks ('bankiershuizen') such as Hope & Co. and Van Eeghen & Co. engaged in underwriting , often forming an underwriting syndicate with a universal bank. The market for underwriting securities was competitive, with different syndicates competing for the same securities to underwrite.

Security issuances were fixed price offerings that the public could subscribe to. Securities were mostly issued around par (see Figure B3). Underpricing was in general limited, unrelated to the volume of the security issuance, but correlated mainly with stock market sentiment (Legierse, 2023). Importantly, the underwriting fee was not a function of the perceived riskiness of the issuance (see Table D1). In the event of oversubscription, the securities would be divided pro rata among the subscribing investors. The demand for newly issued securities in the 1920s stemmed from different groups of investors, depending on the type of issuer. Institutional investors mainly invested their money in Dutch and Dutch East Indies government bonds (Legierse, 2023). The upper class elite of private capital owners dominated the share purchases of industrial firms (Westerhuis and de Jong, 2015; Renooij, 1951). Since the standard nominal value of a security was $f.$ 1,000, few small private investors had pockets deep enough to participate.[12]

We focus on the most common type of underwriting whereby the underwriter would agree to guarantee the placement of securities for a specific price (firm-commitment). This way, any placement risk was shifted from the firm to the bank. The underwriter then tried to sell the securities on the predetermined issuance date. If there was not enough investor demand, the underwriter was obliged to buy the securities at the pre-specified price, which it would try to sell in the future. When an issuance was successful, the underwriter received a commission premium, which was often a few percent of the total issuance amount (on average 3% in our sample). Underwriting was considered successful when subscription on the issuance day was close to the full amount of securities offered. We consider every issuance above 90% subscribed as successful. When the issuance failed, the underwriter was exposed to substantial losses, up to the total underwriting guarantee. The bank could potentially sell the securities for a good price at a later stage, but this was uncertain ex ante and selling later often resulted in losses (Geljon, 2005; Kymmell, 1998). In many cases, unsold underwritten securities remained on the bank's balance

[12] A thousands guilders in the 1920s is equivalent of €15,000-20,000 nowadays, depending on the year in which the security was issued. Source: Central Bureau for Statistics, https://www.cbs.nl/nl-nl/visualisaties/prijzen-toen-en-nu

sheet.[13] For instance, a substantial amount of Chinese government bonds ($f.$ 3.9 million) issued in 1922 were still on the balance sheet of the NHM in 1934 (De Graaf, 2012).

Underwriters often set up a syndicate ("consortium") to share the underwriting risk. Wertheim (1891) defines a syndicate as a temporary collaboration between banks or other financial intermediaries to sell securities to the public. At the end of the syndicate agreement, remaining unsold securities had to be divided among the participants in the syndicate, proportional to their initial underwriting guarantee. The NHM had several underwriting partners with whom it regularly formed syndicates, depending on the type of issuance that was offered.[14] Each member of the syndicate could subsequently offer a sub-participation to other 'friendly relations', such as banks, firms or their executives.[15]

Underwriting offers can be divided into two categories: offers where a bank was one of the lead underwriters, and offers where the bank was a syndicate participant. If the bank was a lead underwriter, it received an underwriting offer from a public or private issuing entity that wanted to raise a certain quantity in the market. The bank—together with any other lead underwriters—would negotiate the issuance price and quantity. If the amount requested by the firm was deemed too large for the leads to underwrite on their own, they could invite other banks to participate. The NHM's guarantee was substantially higher when it was a lead underwriter (see Figure B1). Also, after a failed issuance, its reputational damage would be larger; the prospectus, offered to the public, only included the names of the lead underwriters, not of the participants (Wertheim, 1891).

If the bank was a participant, it could only accept or reject the offer from a lead underwriter who would have first negotiated the terms with the issuer. It was unable to check the issuer's books before issuance. Accepting participations offered by friendly relations was necessary to "prevent isolation on the stock market and consequently be excluded from doing business."[16] Figure B4 shows that on average, issuances where the bank was a participant rather than a lead underwriter were slightly less successful, suggesting that the bank was more careful in selecting issuances to underwrite when its name was on the prospectus.

Investors could subscribe to the issuance through the offices of the bank in the big cities or of smaller affiliated banks in the provinces. As is typical today, allocation was pro rata and we see many instances where investors subscribed for extraordinarily large amounts ("majoreren", or order inflation) because they knew the issuance would be oversubscribed,

[13] End of year reports from the executive board to the supervisory board included lists of unsold securities still held by the bank. Nationaal Archief, Den Haag, De Nederlandsche Handel-Maatschappij (NHM), number 2.20.01, inventory number 54: yearly report 1930.

[14] Amsterdam municipal bonds were underwritten together with Hope & Co. and the Amsterdamsche Bank, while for foreign issuances the bank often collaborated with the German Mendelssohn & Co.

[15] NL-HaNA, NHM, 72, 1907 p. 47-48.

[16] NL-HaNA, NHM, 36: 12-11-1907 - 14-11-1907, authors' translation.

leading to subscription rates of sometimes 40 times the issuance amount. Investors would not observe the exact amount a bank would underwrite.

1.3 Bank management

The NHM had five to six executives ("bankers"): the president, three or four directors and a secretary. Three bankers, president Van Aalst, director Bierens de Haan and secretary Abbing, remain in the same role throughout our period of analysis. The composition and number of directors changed over time: 3 in 1924 and 1930 and 4 in the years 1919–1923 and 1925–1929. All bankers were selected based on their expertise and experience (De Graaf, 2012). The president had the decisive vote in meetings. The bankers were overseen by a Supervisory board.[17]

The bankers received a fixed base salary depending on their position. The president received a f. 20,000 fixed salary in 1919; the directors were paid 80% and the secretary 40% of the president's salary. [18] These fixed salaries were low compared to other banks (De Graaf, 2012). They were increased once in our sample period, by 75% in 1925, when the economy recovered from the banking crisis.

Additionally, the bankers received a bonus ("tantième"), which was a percentage of the bank's profit ranging between 1-2.5% based on the same ratios as the fixed base salary. Figure B2 shows the evolution of the base salaries and bonuses over the years of our sample. With the exception of 1923, the bankers' bonuses were well above their base salaries. At its peak in 1919, the bonuses were almost 14 times higher than the base salary. Much like today, the relatively large size of bonuses likely gave bankers strong incentives to take risk.

Finally, bankers often held supervisory board positions at firms the bank worked with to oversee affairs there. This often generated additional income.

1.4 Personal guarantees in underwriting

The bank's supervisory board was in continuous debate with its bankers about the latter's income. The board was reluctant to increase fixed salaries, as this would decrease the

[17] The supervisory board represented the bank shareholders and had to consist of at least ten members chosen from the electoral committee of the local Chamber of Commerce. The supervisory board members received a small attendance fee and a share of the bank's annual profits (2.5% to be divided among all members, usually around 15 directors). They met as often as was seen fit to audit the bankers. They had the power to accept or reject the balance sheet (including the amount of dividend to be paid out) and the yearly reports of the directors.

[18] The secretary received an additional yearly fixed compensation to increase his share to 50% of the president's salary, but his original base salary was determined at 40%.

bank's profits, especially in meager years.[19] The bankers often complained about the low level of fixed pay.[20]

The supervisory board was also cautious about using tantièmes (bonuses) in fear of excessive risk taking. As one supervisory board member indicated:

> "To view tantièmes in the bank's profit as an incentive for better management is a folly. NHM's bankers are always persons that do not require such an incentive. However, even for an institution as ours, with such substantial capital, there still exists the possibility, that such an incentive could work counterproductive, and could lead to large, even risky speculations, in which the bankers would share the profits, but not the losses."[21]

Upon his appointment as President in 1913, Van Aalst pushed for another structural form of payouts: personal guarantees in security underwriting. Bankers would personally guarantee a fixed part of the issuance over which they would receive the underwriting fee. If the issuance failed, they had to personally purchase a fraction of the unsold securities in proportion to their guarantees. That is, if half the issuance remained unsold and they had guaranteed the placement of 50 shares, they had to personally purchase 25. Market prices for failed issuances were typically (far) below the issuance price, or were not even traded for several months, and this exposed the bankers to potentially large losses.

The use of personal guarantees in security underwriting was well-known among Dutch limited liability banks (Boissevain, 1891). NHM had used them with some frequency before 1913 as a way to underwrite larger amounts that the bank would otherwise have been comfortable with.[22]

Van Aalst, who was a powerful figure within the bank (De Graaf, 2012)[23], made these personal guarantees a standard element of the bank's underwriting. It seems that this was a way to increase the bankers' pay that was acceptable to the supervisory board. Personal guarantees did not reduce the banks' profits in a way fixed salary payments would; at least

[19] NL-HaNA, NHM, 37: 7-5-1937.

[20] For example, comments were raised about the "improper ratio between salary and tantième against which [Secretary Abbing] and the bankers appointed after him repeatedly voice their objections." NL-HaNA, NHM, 4129: 24-1938, authors' translation.

[21] NL-HaNA, NHM, 33: 21-11-1879, authors' translation.

[22] For example, in early 1911, bankers provided guarantees in the issuance of a 4% bond of a colonial company because "it was not deemed desirable that the NHM would participate for more than $f.$ 100,000 in the syndicate." Two weeks later, they also provided guarantees in an IPO of a newly established company since "the exposure for the bank would have been too high otherwise". NL-HaNA, NHM, 437: 30-01-1911, 10-02-1911, authors' translation

[23] According to banker Ernst Heldring, who would become president of the bank in 1939, Van Aalst had great power within the bank and the Netherlands, and only surrounded himself by followers and rather dependent businessmen. See Ernst Heldring (1970), *Herinneringen en dagboek*, e.g. p. 480, 530, 556, 623.

in theory, larger personal guarantees did not reduce the bank's profits. At the same time, the downside exposure inherent in the personal guarantees did not incentivize excessive risk taking the way tantièmes might. By the 1920s, personal guarantees in underwriting had become common practice among other Dutch limited liability banks as well (Van der Werf, 2014; De Graaf, 2012): in the confidential meeting minutes we observe the bank extending participations to boards of other banks.

Guarantees were assigned to all bankers, regardless of their attendance at the meeting deciding on a specific security issuance. It was clear to everyone involved that these personal guarantees exposed the bankers to substantial downside risk. According to Boissevain (1891), even the most experienced and skilled banker could not be certain of the success of a security issuance. The guarantees were substantial compared to the bankers fixed annual salary. The median (average) personal guarantee per issuance during our period of analysis was $f.$ 17,500 (26,000) for the president, $f.$ 14,000 (20,000) for each director and $f.$ 10,000 (14,000) for the secretary. One average underwriting guarantee was comparable to the entire annual fixed salary. Some in the supervisory board considered the personal guarantees "not so much as a source of profit, but primarily as a risk".[24] One of the NHM's bankers, Secretary Abbing, suffered substantial losses from personal guarantees provided in the 1920s resulting in a debt due to the bank of $f.$ 60,000.[25]

After Van Aalst resigned as President in 1934, the use of personal guarantees virtually disappeared. It appears that Van Aalst had been relatively risk-tolerant and willing to face substantial risk to earn additional income. The system had started to break down after the onset of the Great Depression. By 1931, only 6 of 24 underwriting instances were personally guaranteed by bankers (down from 30 of 49 in 1930). After Van Aalst's resignation, there was renewed debate in the supervisory board about the costs and benefits of the personal guarantees. There was a proposal to deposit the fees from personal underwriting into an escrow account to cover future losses (to avoid situations like Abbing's).[26] In the end, the system was abolished in 1937 at the same time that the underwriting business was dormant and bankers' fixed salaries were increased.[27]

The details about the bankers' personal guarantees were kept secret from the public. We only observe the size of the bankers' guarantees because the bank's secret minutes have been made available to the public. The prospectus of each security issuance contained only information on the banks that were involved as underwriters (without revealing the

[24]The meeting minutes initially spoke of "sacrifice" rather than "risk". NL-HaNA, NHM, 37: 27-11-1936

[25]NL-HaNA, NHM, 4129; 37: 24-11-1937.

[26]"Bankers should only be allowed to participate in syndicates if they would form a 'consortium' between themselves that should include paid-in capital, yearly balance sheets and profit and loss statements and which could pay out dividends to the bankers, only after securing sufficient retained earnings". NL-HaNA, NHM, 37: 29-05-1934, authors' translation.

[27]NL-HaNA, NHM, 37: 27-11-1936.

exact exposure each bank had). Even if bankers revealed their exposure to others *ex ante*, the signal was not verifiable *ex post*. The secret minutes were only made available gradually to researchers between 1967 and 2010. It therefore seems unlikely that bankers could influence the outcome of the security issuance by signaling their exact guarantees to investors.

2 Data description

2.1 Sources and data construction

We hand-collect data from the archival records of the NHM. The bank is the only large Dutch banks with nearly complete detailed records available at the Dutch National Archives. Decisions regarding bank underwriting amounts and personal banker guarantees are documented in detail in minutes of confidential executive meetings ("secret minutes"), annual reports and security books. We hand-collect this data for all underwriting decisions from 1919 to 1930. We start in 1919, right after WWI when the number of issuances on the Dutch stock market grew substantially. We exclude the period after 1930 due to the lower number of underwriting opportunities and personal guarantees.

We have information on the size bank's underwriting position, the underwriting fee, security and issuer characteristics, the currency of the issuance, other syndicate members, sub-participants, (supervisory) board positions of the bankers at the issuer and previous lending relations and the subscription rate of investors. We analyze the data in the original currency, to keep round numbers of both the bank's underwriting position and the bankers' guarantees. We only scale Austrian Kronas (multiplied by 0.001) and British Pounds (multiplied by 10) to approximate their Dutch guilders value and avoid outliers. We rely on De Graaf (2012) and issuance prospectuses for the bankers' directorates at the issuing firms and for information on previous interactions with issuers.

Second, we retrieve additional data on the issuers' underlying fundamentals from Van Oss' yearbooks and prospectuses over the same period 1919-1930. We hand-collect company balance sheet data and (local) government financial information. For firms, we hand-collect total debt, equity, assets, capital, age of the firm, profitability, retained earnings and dividends. We try to resemble as much as possible the information that investors had available at the time of issuance by using balance sheets and prospectuses from the year before the issuance. If this information is not available due to the issuance taking place early in the year, we use information from two years before. We use the balance sheet of the underlying firm if it is part of a holding company (if available). We do not need to convert the firm balance sheet items to guilders, as we only use financial ratios

and scale variables by total assets. We use the exchange rate from prospectuses, or, if not available, the yearly guide to stock market prices ("Gids bij de Prijscourant") to convert the debt and deficit numbers into guilders.

Third, we use share prices to calculate abnormal returns on issued securities. That is, we compute the buy-and-hold abnormal returns for 1-12 months relative to the issue price. We hand-collect monthly security prices from the so-called *Tabellenboeken*, where we take the midpoint of the monthly reported lowest and highest prices. For years where the Tabellenboeken do not have monthly high and low prices (1919, 1920 and 1924), we consult the daily *Prijscourant*, where we take the end of month prices (or second to last trading day if the last day is missing). We construct a value-weighted monthly share index using the share prices of the largest listed firms reported in *Economisch Statistische Berichten*.[28] To calculate abnormal returns for bonds, we hand-collect the returns on the Dutch on-the-run long-term government bond from the *Tabellenboeken*. Appendix A explains in more detail the data collection for the returns sample.

2.2 Predetermined step function for personal guarantees

The size of the banker's personal guarantees depended on the bank's underwriting amount. There existed a discontinuous step function where the personal guarantees increased step-by-step at predetermined discontinuities. Figure 1 plots the observations in our main sample where the bank is one of the lead underwriters. The personal guarantees are always a round number. The figure shows that the majority of observations follows a step function (although there are deviations). We show the median guarantee among all directors to deal with cases in which one of the directors had a smaller or larger guarantee. In line with the fixed salaries and tantièmes, the typical guarantees of the president and secretary were 125 and 75%, respectively, of the directors' guarantee (see Figure B2). [29]

The archival documents do not contain any explicit information on the exact rule that the banker followed, yet Figure B5 shows a clear pattern in the form of a step function. We infer the exact steps of the rule both manually and via a dynamic programme (see figure F2) from the step function. Using the fixed salary ratios and without taking board connections into account, steps that determine the rule are subject to the following restrictions: First, the steps cannot overlap. Second, the steps should be monotonic: banker guarantees should increase for each step (*monotonicity*). Third, the range of the bank's underwriting position should weakly increase with each step (*widening steps*). Fourth,

[28] We thank Elisa de Weerd for sharing the weekly historical ESB issuances.

[29] The most frequent ratio of the president's (secretary's) guarantee to the directors' is exactly 1.25 (0.75): in 77% of issuances in 1925–1930 and about half in 1919–1925. In rare cases where all directors appear to deviate from that ratio, we adjust the median director guarantee with 80% of the president's guarantee.

board connections should not determine the rule (*no interlocks*). Board connections can lead to deviations from the rule as bankers might have more private information and/or private benefits. While we control for this in our regression analysis, the observations cannot be relied upon to determine the rule, and we leave them out.[30] Fifth, each step contains at least one issue whose median director guarantee lies exactly on the step's level (*admissible levels*), and these levels are even numbers that the bankers used at least twice in the period 1925–1930 (*anchored steps*).[31] Sixth, below 150,000 the bankers do not underwrite, simply because the exposure for the bank is too small for personal guarantees (*floor*; see Figure F1). We exclude issuances where the bank's underwriting amount exceeded 5 million: there are too few to back out the rule. Between 150,000 and 5 million our rule has seven steps in 1919–1925 and eleven in 1925–1930.

There was a shift in the rule after October 1925, with bankers generally having lower guarantees. Potential reasons for a shift in the rule are the banking crisis of 1921-1924, several changes in the composition of the executive board, and the increase in fixed pay in 1925. The bankers' personal income had substantially declined during the crisis period of 1921-1924, with bonuses in some years approaching the low base salary (see Figure B2). Second, in 1925 the bank hired a younger additional board member that initially only participated for half of the director's guarantees. In the same year, the supervisory board approved an increase in the bankers' fixed base salary. This appears to have gone hand in hand with lower personal guarantees in underwriting such that bankers would have less of their personal wealth at risk. Section 4.4 shows that an exhaustive search under these restrictions recovers our rule in both periods, and that data without a rule cannot reproduce its compliance.

In our empirical analysis, we compare issuances above and below the rule's discontinuity. In particular, we include bin fixed effects in our regressions. Each bin starts in the middle of a step, and ends in the middle of the next step. Figure 2 has a visual example. We exclude observations with bank underwriting amounts less than f. 50,000 which never have any banker guarantees.[32] Using these bins, we can compare observations where the bank had a similar position, keeping issuance size largely constant.

[30]The exception is 1919–1925 above 2 million, where three issuances with an interlocked directorate do follow the rule (Section 4.2).

[31]In 1925–1930 every step contains at least two issues on its level; in 1919–1925 issues are too sparse for that, so we take the levels from 1925–1930 and let a 1919–1925 step rest on a single issue . Table F1 shows that the results of the main specification do not change if we use a different method of selecting guarantee levels.

[32]Including this 'zero bin', which contains 23 observations, does not affect the results. 12 issuances (70 observations) fall exactly on the midpoint of a bin. We assign those to the lower bin; only after the round number, the predicted banker guarantee increases. Table E1 shows the results when assigning the observations that fall exactly on a midpoint of a bin to the upper bin instead, or when excluding those observations.

Despite the within-bin comparison, there are still two ways in which the bankers could avoid exposure, making it more complicated to test the effectiveness of the rule: 1) the rule appears to have not been strictly binding as the bankers sometimes underwrote more or less than predicted and 2) the bankers manipulated the bank's guarantee around a threshold to avoid ending up below a discontinuity. We discuss both confounders and how we address them in the empirical design below.

First, bankers did not always adhere to the rule exactly. Deviations occurred more often with domestic securities, especially when the bank had previously done underwriting for the issuer before or when one of the bankers had an interlocked directorate. It is possible that deviations from the rule are correlated with issuance success (e.g. bankers guaranteeing more (less) than required for less (more) risky issuances). Indeed, figure 6 shows suggestive evidence that bankers' deviations from the rule are correlated with the subscription rate of the issuance. To partly address this problem, we implement an intent-to-treat design where we instrument the actual bankers' guarantee with the guarantee predicted by the rule. That is, in the second stage we estimate an approximate local average treatment effect of the compliers, where more weight is put on the observations that follow the rule.

Another way to avoid too much exposure to risky issuances was to exactly follow the rule, but manipulate it. The bank guarantee was likely negotiated rather than assigned, so a natural concern is that the bankers adjusted the bank's underwriting amount to fall just below a discontinuity if they deemed the issuance to be risky.[33] An alternative explanation could be that observations above a discontinuity were issuers with better bargaining power, but it is unlikely that issuers would know both about the bank's gross and net position, and the discontinuities in the rule for the bankers. Moreover, it is difficult to disentangle proxies for issuer bargaining power (firm size, repeated issuance) from issuer quality ex ante. Figure 3 plots the distribution of the bank guarantees for both periods together, where each issue is normalized to its nearest discontinuity. That is, each issuance is scaled by half the width of the tier, which is the gap between two discontinuities. While we do observe slight bunching at or just below the discontinuity (26 out of 238 (11%), it is difficult to separate that from a preference for round-number guarantees (20 of those 26 observations). Every discontinuity is itself a multiple of f50,000, and so are 70 of 239 observations overall (29%).

Figure 4 addresses the round-number preference. It plots the distribution of bank guarantees if we consider every 50,000 multiple to be a placebo threshold (excluding the 61 observations around thresholds that are actually part of the rule). The figure shows that

[33]The bank could either extend additional participations to other banks or negotiate a lower underwriting amount with the issuer, such that in both cases the net bank guarantee, which determined the personal guarantees, would be lower.

bunching occurred often in multiples of 50,000, with 53 of 178 observations (30%) located in the bin just before a threshold (51 exactly on the threshold). Note that the shares are not directly comparable: bin width differs in the counterfactual graph and it contains 90 rather than 7 or 11 thresholds. Still, of the 61 issues whose nearest round number is a discontinuity, 19 sit exactly on it (31%); of the 178 whose nearest round number is not, 51 sit exactly on it (29%). A round number attracts issues no more strongly for being a discontinuity. Exploiting the rule revision of October 1925, which moved the discontinuities while leaving the set of salient round numbers unchanged, we find that a round number attracts 1.28 times the mass in the regime in which it is a discontinuity (95% CI 0.70 to 2.33; exact conditional p = 0.36). With only 19 observations exactly on a threshold, the test can detect excess bunching only if it is 2.4 times larger than the mass for ordinary round numbers. We can therefore not reject the hypothesis that a discontinuity does not attract more mass than an ordinary round number. Adjusting the bank's underwriting amount is in any case only one margin along which bankers could avoid larger, riskier exposure. Our results point to another and, we believe, more important response: with more of their own capital at risk, bankers underwrite better issues of a given size.

2.3 Summary statistics

The bank is active in underwriting 561 security issuances from 1919-1930. We exclude observations where the exact bank underwriting position or the success of the issuance is missing. 318 of the 561 issuances are participations, i.e. cases where the bank is not a lead underwriter but a member of the syndicate (64) or a sub-participant (254). We exclude an additional four securities where the bank guarantee is too low (below $f.$ 50,000) for the bankers to underwrite. Our main analysis contains the remaining 239 security issuances where the bank is one of the leaders of the syndicate. Using all the bankers in our sample as individual observations provides us with 1,391 observations. In our sample, 215 securities (1,255 observations) have personal banker guarantees.

Table 1 shows the characteristics of the security issuances NHM underwrote by year. During the most severe banking crisis years of 1922-1924 there were hardly any: 1922 was the least active year with only three issuances. In the years after the banking crisis, the number of issuances increased to a peak of 42 in 1928. The other columns in the table show the yearly percentage of issuances that were stocks (and IPOs) or (local) government issuances, had foreign issuers, were issued in Dutch guilders, were rights issues or preferred issues and were successful. Rights issues were exclusively for existing shareholders, so there was no scope for oversubscription. Preferred issues gave preferential treatment to existing shareholders, but were afterwards open to the general public. From 1919-1920, in the post WWI years, the Amsterdam stock exchange saw a boom in the issuance of

domestic stocks. More than 60% of the securities in our sample are stocks, all issued in Dutch guilders. Moreover, 100% of the securities are from domestic issuers in 1919 and 89% in 1920. In the other years of our sample, the majority of the observations are bond issuances. Especially after 1924, the share of foreign security issuances dominates, even though a substantial number is still issued in the Dutch guilders. Foreign firms and governments often issued securities in multiple markets, where cross-border syndicates were set up and securities were listed on several exchanges (especially for very large issuances). The securities listed on the Amsterdam Stock Exchange were sometimes issued in guilders in those cases. Issuance success varies over time, with a majority of securities fully subscribed in all years but 1922, 1928 and 1929. The increasing number of relatively low quality issuances (i.e. not fully subscribed) in 1928 and 1929 is an indication of the overheated stock market.

Table 2 shows the summary statistics for our sample of security issuances underwritten by NHM. The observations are mostly in guilders (59%) and dollars (28%). Realized banker guarantees, around 20,000 in the currency of issuance, are slightly smaller than predicted by the rule, largely because in some cases a banker did not participate in the guarantees. The bank's average share in the underwriting syndicate was 34%. Of all issuances, 59% were successful (had a subscription rate above 90%), with 49% of issuances fully or over-subscribed. The distribution of subscription rates is highly skewed. While the median issuance is fully subscribed, the average is 25 times oversubscribed. This is likely driven by order inflation: investors would put in extra large subscriptions if they expected the issuance to be oversubscribed. While the costs for the bank associated with under-subscription are linear in the subscription rate, benefits are capped at full subscription. We therefore take the log of the subscription rate, where zero indicates a full subscription, which better reflects the upside and downside of underwriting (see Figure 5 for the distribution).

Only 3% of issuances in our sample are IPOs, with SEOs and bond issuances making up 18% and 79% , respectively. In 17% of cases, the issuer has a borrowing relationship with the bank; in 15% of cases there is an interlocked directorate between the issuer and the bank. These two variables are not collinear: only half of the observations where the issuer has a borrowing relationship with the bank also feature an interlocked directorate.

The other variables in Table 2 come from firm balance sheets and prospectuses, all in the year before issuance. This information is not available for the entire regression sample. Issuance/size is the total security issuance divided by the total assets of the firm. On average, firms issued 13% of their assets in additional capital or debt.On average, retained earnings were 10% of assets, with profits 5%. The average firm in our sample was 25 years old, and had twice as much debt as equity on its balance sheet. Debt makes up nearly

half of total assets. Only a few (local) governments provide information on outstanding debt and budget surplus or deficit in the year before issuance. We exclude those from the analysis on ex ante riskiness. Outstanding debt is on average $f.$ 228.59 per capita, while budget deficits are $f.$ 0.32 per capita.

Table 3 compares observations within bins where the bank's underwriting amount is just above or below a discontinuity in the rule. By design, the amount is higher above the discontinuity and this difference is statistically significant. The average actual and predicted personal guarantees are $f.$ 5,000 and $f.$ 6,500 higher. The difference between the two is due to the deviations from the rule. The dependent variable, the log subscription rate, is also significantly higher and the probability of an issuance being fully subscribed increases from 43% to 55%.

Other variables do not significantly differ around the discontinuities. Above the discontinuity, the underwritten securities are slightly more likely to be bonds and issued in guilders. The underwriting fee that bankers received is even on average identical above and below the discontinuity. The number of bankers in our sample varies between 5 and 6 depending on the year. Below the discontinuities, the number of bankers participating in underwriting is slightly above 5. This is due to more issuances where the bankers did not participate at all. Again, this difference is not significant.

3 Empirical approach

We examine the relation between personal guarantees and the quality of security issuances through investors' subscription rate: the lower the subscription rate, the higher the risk for the bank that it would need to hold securities on its balance sheet at a loss. The step-function led to increases in banker guarantees around predefined discontinuities, but Figure 6 shows that bankers sometimes deviated from this rule. It is possible that these deviations are correlated with the ex ante likelihood of an issuance succeeding. Figure 6 provides suggestive evidence for this: actual personal guarantees are more likely to exceed the predicted personal guarantees determined by the step function for issuances that were successful ex post. To address this concern, we use an intent-to-treat design, instrumenting the actual personal guarantee with the predicted guarantee implied by the step function using 2SLS. This approach isolates the exogenous variation in personal guarantees driven by the step function, allowing us to estimate its effect on the quality of the issuance.

The first stage estimates the relationship between the actual personal guarantee and the instrument, the predicted personal guarantee, as follows:

$$D_i = \beta Z_{i,d} + \Gamma' X_i + \eta_t + \zeta_b + \kappa_d + \tau_g + \epsilon_i \quad (1)$$

where D_i is the actual and $Z_{i,d}$ the predicted banker guarantee in thousands of the currency of issuance. Vector X_i includes a set of controls: the share of the bank guarantee in the syndicate and dummies for 1) a stock or bond issuance, 2) a public or private issuer, 3) a foreign or domestic issuance, 4) an interlocked directorate, 5) an IPO, 6) an existing bank borrower relationship and 7) a rights issue (issuance exclusively for existing shareholders). Importantly, the share of the bank in the syndicate is based on the gross rather than the net bank guarantee (which determines the bankers' guarantees), such that it does not affect our treatment variable. We include fixed effects for year η_t, banker κ_d, industry τ_g and importantly, bin ζ_b. With bin fixed effects, we compare issuances above and below a discontinuity. This also controls flexibly for bank guarantee size. With year fixed effects, we compare issuances within each year to avoid that the post-war boom and banking crisis period affect our results. With industry fixed effects, we control for heterogeneity in security issuances, such as industry-specific risks and capital needs. With banker fixed effects, we control for changes in executive composition. We cluster standard errors by issuance.

In the second stage, we regress the quality of the security issuance i on the instrumented personal guarantee:

$$Y_i = \beta \hat{D}_{i,d} + \Gamma' X_i + \eta_t + \zeta_b + \kappa_d + \tau_g + \tilde{\epsilon}_i \quad (2)$$

For our main regression analysis, the outcome variable Y_i is measured as the logarithm of the subscription rate. The independent variable of interest is $\hat{D}_{i,d}$, the banker guarantee instrumented with $Z_{i,d}$.

We estimate the same regression specifications for president Van Aalst, director Bierens de Haan and secretary Abbing individually. These three bankers remained in their function throughout our period of analysis. We run separate regressions for our sample of firm issuances, where we also consider different dimensions of firm risk.

4 Results

4.1 Issuance subscription

Table 4 shows that issuances with higher banker personal guarantees were substantially more successful. Column (1) to (4) include all observations where the bank is one of

the lead underwriters. Column (5) includes only observations where the bank is the first lead of the syndicate. The top panel displays the OLS results. Personal guarantees are associated with higher subscription rates and therefore higher quality issuances. Column (1) shows that an additional 1,000 (guilders, dollars, etc.) of personal guarantee, within each bin and year, is associated with a 2.7% higher subscription. The results are almost identical when we include industry fixed effects (column 2) and the coefficients increase slightly when we add banker fixed effects (column 3). When including all fixed effects and controlling for the characteristics of the issuance, which is our most conservative estimate (column 4), the coefficient increases to 0.032, which translates to a 21% higher subscription rate.

The OLS results might be driven by cases where the bankers deviate from the rule for higher quality issuances. We therefore instrument the actual banker guarantees with the predicted guarantees according to the rule we back out from the step function. The reduced form results in the second panel confirm that subscription rates are higher when the rule stipulates higher personal guarantees. The coefficients in columns (1) through (4) remain significant but are slightly lower than in the OLS, ranging between 1.5 and 2% higher subscription rates.

The first stage estimates in the bottom panel confirm the relevance and strength of the instrument; it is highly significant and stable in all four columns with a coefficient of around 0.8. The Kleibergen-Paap F-statistic is above 50 in all four columns.

The second stage IV results in the third panel are statistically significant and in line with the OLS estimates. An additional 1,000 in instrumented personal guarantees is associated with a higher subscription rate of 2.5% in our most conservative estimate (column 4). This is somewhat lower than the OLS, suggesting that when bankers deviated from the rule, this was at least in part driven by their ex ante expectations of the issuance's success. The average (median) increase in personal guarantees at a discontinuity is 6,000 (3,000). Our estimates suggest this led to an increase in the subscription rate of about 16.2% (7.8%). Compared to the median subscription rate of 100%, this effect is substantial.

Even though the discontinuities at which personal guarantees increased were predetermined, the eventual size of the issuance was not. One interpretation of our results is therefore that bankers only increased their own and the bank's exposure for higher quality issuances. We explore this further in Column 5 of Table 4 where we only consider cases where the NHM was the first lead (first named) of the underwriting syndicate and likely had more influence over issuance size.[34] The bank is the first lead in 71% of our observations. The OLS results are similar to the full sample, but the coefficients of the reduced form and IV estimates (both significant at 5%) are substantially larger than

[34]NL-HaNA, NHM, 5956.

the full sample results. When the bank is the first lead, an additional 1,000 of personal guarantees leads to bankers underwriting securities that have an approximate 4% higher subscription rate. With an average increase at a discontinuity of 6,000, this means a 27.1% higher subscription rate; almost twice the size as for our full sample, though this difference is not statistically significant ($t = -0.45$).

Table 6 replicates the same analysis for the three individual bankers that were present throughout our sample period. The upside of this alternative approach is that we no longer need to cluster our standard errors. The downside is that we have less statistical power in the first stage: individual deviations from the rule become quantitatively more important compared to our baseline where we consider all bankers together and idiosyncratic deviations largely cancel out. The results in Table 6 are similar to the full sample analysis and even increase in magnitude for the director and secretary in the OLS. The first stage remains statistically significant, but, as expected, the F-statistics are substantially lower between 12 and 16. The IV coefficient estimates are comparable to our baseline, but, due to lower statistical power, only statistically significant at the 10% level. We therefore prefer our baseline estimates where we pool all bankers together.

4.2 Marginal effects

The exposure for bankers was in some cases extreme: if the bank would underwrite more than $f.$ 2 million (or 2 million in another currency of issuance), the bankers would be on the hook for more than their annual salary, just for one security issuance. This is the case for 33 security issuances, of which 29 have personal guarantees. Out of the four bank guarantees without personal guarantees, only one issuance is successful. Almost all of these large deals with personal guarantees that are not successful (subscription ¡ 90%) are domestic securities of well-known and safe institutions (such as municipal bonds of Amsterdam and corporate bonds of Philips) with only two foreign securities, one of them a stock issuance. Interestingly, on average the rule is followed more strictly (72% for large deals vs 52% for deals below $f.$ 2 million).[35] Eight issuances deviate from the predicted rules, including two foreign issuances that turn out to perform the worst: they have a subscription rate below 60%. For both the bond (Societe Metallurgique de Sambre et Moselle SA) and the stock (Internationale Gesellschaft für Chemische Unternehmungen AG), the bankers underwrote less than 60% of what the rule prescribed.

[35]Above $f.$ 2 million, the rule is followed for issuances with an interlocked directorate: six out of seven follow the rule, compared to 6 compliers out of 28 interlocked directorates below $f.$ 2 million. We use this fact to include observations with interlocked directorates in determining the rule for the largest steps in 1919-1925.

After a certain level of exposure, additional liability might be counterproductive. Table 5 shows the regression results without the largest deals. The OLS coefficient declines slightly in magnitude, but the reduced form and second stage coefficients are three times larger, suggesting that the rule was especially effective for deals where the additional liability was limited. The F-stat of the first stage is smaller, but still sufficiently large. The interaction with an above 2 million dummy in column (2) is negative and significant in both the RF and IV. Below *f*. 2 million, an additional 1,000 of guarantee raises the subscription rate by about 8%, above it by less than 2%. This implies that the personal guarantees became much less effective for larger deals.

Figure 7 shows the marginal effects of banker guarantees on the log-subscription rate using the specification from Table 4 adding a quadratic term for actual, predicted and instrumented banker guarantees. The figure shows that increasing banker guarantees at already high levels of guarantees is less effective at improving issuance quality. This is consistent with bankers becoming increasingly averse to taking risk as their exposure grows in size, reducing the marginal effect of even more exposure. This suggests that imposing additional liability on bankers will only be effective if their downside exposure is limited to begin with.

4.3 Mechanism

Higher subscription rates for issuances with larger bank director guarantees can arise for three non-mutually exclusive reasons: (1) directors select issuances with less downside risk, (2) directors underprice the security to improve the probability that the issuance is a success, or (3) they exert greater effort in placing securities. In this subsection, we provide evidence consistent with mechanisms (1) and (2) by demonstrating that issuances with larger director guarantees exhibit stronger fundamentals and generate higher subsequent returns. While we cannot empirically test mechanism (3), the confidentiality of the personal guarantees rule out that signaling was part of the effort exertion by the directors.

4.3.1 Ex ante riskiness

First, we compare the security issuances' ex ante quality and riskiness.[36] For the subset of firm issuances, we consider the age of the firm, with older firms likely having safer issuances because of the better information availability for investors (e.g. Legierse 2023). Further, we consider leverage, and current profits and retained earnings over assets. Focusing on firms shrinks our sample to 793 observations covering 137 security issuances.

[36]For the sample of public institutions, we lack power with only 55 issuances (324 observations) with risk measures available. We therefore focus only on firms in this section.

We compare the distribution of ex ante risk measures between successful and unsuccessful issuances. Figure 8 shows the kernel densities. Firms with successful issuances are relatively older and have less leverage, more retained earnings and higher reserves on average.

Table 7 implements our regression analysis for firm security issuances. Column (1) considers the log subscription rate. The coefficients in the OLS, RF and IV specifications and the F-statistic in the first stage are all larger than for the main sample. For this subsample, the average increase in the bankers' guarantees at a discontinuity was 7,000. According to our point estimates, this was associated with an increase in the subscription rate of about 28% in the IV.

Columns (2) and (3) look at firm risk. We take the first two principal components (PCs) of the variables in Figure 8, log(age), debt over equity, and retained earnings and current profits over assets, as well as the size of the issuance over assets and the fraction of short term over total debt. Table C1 in the appendix has details. The first PC loads more heavily on balance sheet health, while the second PC loads on firms' maturity. Positive values indicate safer and more mature firms. Together, the first two PCs explain more than 50% of the overall variation. Table 7 shows that both PCs are higher if underwriting guarantees are above the threshold, but the effect is only marginally statistically significant for the first principal component. These results are consistent with higher guarantees pushing bankers towards underwriting safer security issuances.

These results suggest that security issuances above a threshold exhibited better *ex ante* fundamentals. This improved the probability that an issuance would be a success, as evidenced by the higher subscription rates.

4.3.2 Returns and underpricing

Next, we compare ex-post security returns for issuances above and below an underwriting discontinuity. We consider buy-and-hold abnormal returns. For equities, we construct our own stock market index; for bonds we construct a return index for Dutch on-the-run government bonds. We consider abnormal returns over the subsequent one (1m), two (2m) six (6m) or twelve months (12m). Figure D1 shows the distributions by success of the issuance (with a subscription rate higher than 90%). Successful issuances are correlated with higher returns from the first month onward, suggesting that underpricing improved the chances a security issuance would be a success. Two months after issuance, unsuccessful issuances are still correlated with lower returns. For six and twelve months the successful issuances still have higher returns on average, but the tails become much

larger and the point estimates lower and less reliable. For precision, we focus on the first month after issuance in the regression analysis.

Regression results are in Table 8. Column (1) considers how frequently newly issued securities were traded – absence of trade can indicate problems. Higher banker guarantees increase the months a security is traded, suggesting that bankers selected more liquid securities to underwrite.

In columns (3)–(5) we account for the missing prices of those less liquid securities by imputing the return with a percentile of distributions of all stocks/bonds within 3 years of the respective price date. Column (6) is not adjusted but includes only securities that have a return for every month. For every measure of BHAR, securities with underwriting guarantees above the threshold had higher abnormal returns in the first month after issuance. A $f.1,000$ larger banker guarantees leads to an abnormal return that is 0.07-0.15 percentage points higher. Taking the average jump in banker guarantees at a threshold ($f.6,000$), this translates into a higher abnormal return of around 0.5-0.9 percentage points. The sample average BHAR for one month (unadjusted) is 3.06 percentage points, so this is a substantial effect. These findings suggest that for issuances above a discontinuity, bankers were more likely to underprice securities to improve the chances of success. Figure E1 shows that the results are consistent when using different imputation methods, and are only significant at the first month after issuance. We find no statistically significant difference in BHAR for six and twelve months after issuance.

4.4 Validating an exogenous step-based rule

The archival documents do not state the rule: we infer its steps from the pattern of observed guarantees, and we cannot be certain that the step function we infer is the one the bank's managers followed. We therefore proceed in three steps. We first show that a rule exists. Then we show that our step function is not a choice among many that fit the data equally well: under the restrictions of Section 2.2, an exhaustive search returns our step function. Finally, we provide evidence that the estimates do not depend on a particular choice among plausible restrictions.

A rule exists Under the null hypothesis, guarantees are set ad hoc: a smooth step function in the bank's underwriting amount plus noise, with round-number preference. We draw 10,000 counterfactual datasets per period from this null (estimated on our own data, with heteroskedastic errors; see Appendix F). For every counterfactual dataset we run the same dynamic programme under the same restrictions and record the highest share of issues that lies exactly on any admissible step function. Without a rule, this share is 36% on average in 1919–1925 and 32% in 1925–1930, against 54% and 60% on

our step function in the real data (Figure F3). The null is rejected with $p = 0.005$ for the first and $p = 0.0001$ for the second period: 45 of 10,000 counterfactual datasets reach our compliance in 1919–1925, and none in 1925–1930. Intuitively, exact compliance is the fingerprint of a rule: ad-hoc rounding produces guarantees that lie *near* any reasonable step function, but it rarely places more than half of them *exactly on* one specific step function.

The rule is the best-fitting rule Next, we search over all step functions that satisfy our restrictions, with breakpoints on a grid of round $f.$ 50,000 amounts, and select the one on which the most issues lie exactly. The search is a dynamic programme that evaluates every admissible step function and returns the global optimum (Appendix F). It uses the same data as our rule: issues without an interlocked director, and for 1919–1925 the two issues above $f.$ 3.5 million and one issue above $f.$ 2 million in which all directors had the same personal guarantee. Figure F2 plots both the step functions from the manual search and dynamic programme. In both periods the search recovers our step function, with seven (1919–1925) and eleven (1925–1930) steps below $f.$ 5 million.[37] In 1919–1925 its prediction coincides with the rule used in the paper for all 35 issues, and in 1925–1930 for 145 of 146 issues; the remaining differences are breakpoints that fall in ranges without any issue. No admissible step function places more issues exactly on the rule: 54% in 1919–1925 and 60% in 1925–1930.

The rule is not cherry-picked The rule rests on the restrictions of Section 2.2, and a different set of restrictions could yield a different step function and different estimates. Tables F1, F2 and F3 therefore re-estimate our main specification (Table 4, column 4) with the rule the dynamic programme selects when we change the restrictions, each time with bin fixed effects built around that rule's own thresholds. Table F2 shows that when we apply all restrictions, the selected rule in the dynamic programme is identical in levels, but differs in bin range. The IV estimate is very comparable at 0.028 (s.e. 0.012), against 0.025 (0.012) with our own bins. Table F3 shows that the results disappear when cumulatively dropping restrictions once the increasing bin width is dropped (*widening steps*). When leaving one assumption out (Table F2), we observe that non-increasing bin widths (*widening steps*), interlocked directorates included (*no interlocks*), and increasing guarantees (*monotonicity*) are key for our results. Each of these restrictions is reasonable and required for internal consistency. Bin width should weakly widen: in both periods the upper steps are visibly wider than the lower ones (Figure B5). Issues with an interlocked directorate should not determine the rule, since it affects whether the bankers deviate

[37] While the levels are identical with the dynamic programme, the bin boundaries vary slightly. The second row of Table F3 shows that the results are robust for using the dynamic programme boundaries, which give even slightly higher coefficients.

from it (Figure 6). Banker guarantee levels should rise with the bank's guarantee to ensure symmetric risk-sharing.

5 Conclusion

This paper analyzes the effect of bankers' personal guarantees on bank risk-taking. We exploit uniquely detailed data from a large and prestigious Dutch universal bank in the 1920s in which bankers had to provide personal guarantees for their bank's underwriting activities. Underwriting was risky at the time, with 50% of issuances under-subscribed. Bankers chose whether and how much the bank would underwrite and directly influenced how much risk the bank took on. They faced additional liability as they had to personally guarantee a pre-determined part of the issuance. This guarantee followed a step function in the bank's underwriting amount: at pre-specified discontinuities, the bankers' personal guarantees increased by a discrete amount.

Our results suggest that additional liability, directly tied to bankers' personal decisions, can be a valuable tool in decreasing bank risk-taking. We use the subscription rate as an ex post measure of issuance quality and risk. Issuances right above a discontinuity, for which bankers had to provide larger guarantees, had substantially higher subscription rates. Issuances above a discontinuity featured firms with healthier balance sheets. They were also followed by higher buy-and-hold-returns in the first month after issuance. These results suggest that bankers were more selective in what securities they would underwrite and insisted on a more conservative pricing if they had to provide larger guarantees. The effectiveness of these guarantees declined with size. At high levels of guarantees, bankers already limited the riskiness of issuances, and the marginal effect of extra guarantees was smaller. This suggests that imposing additional liability on bankers will only be effective if their downside exposure is limited to begin with.

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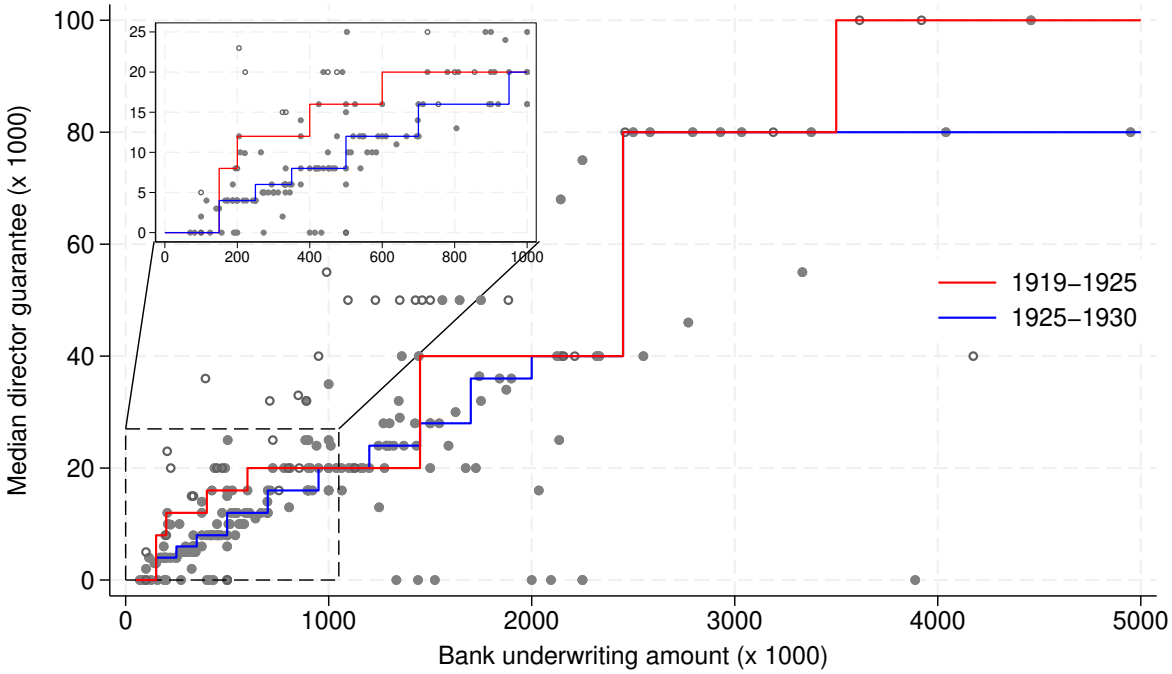
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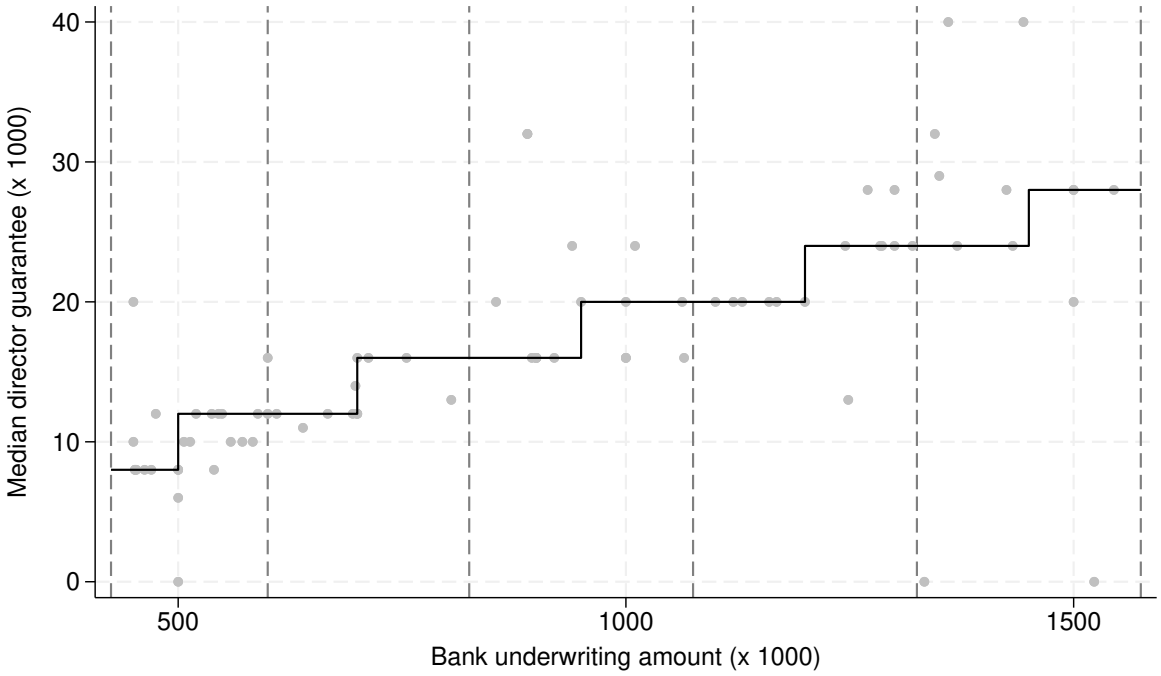
Figures

Figure 1: Bank and bankers' guarantees



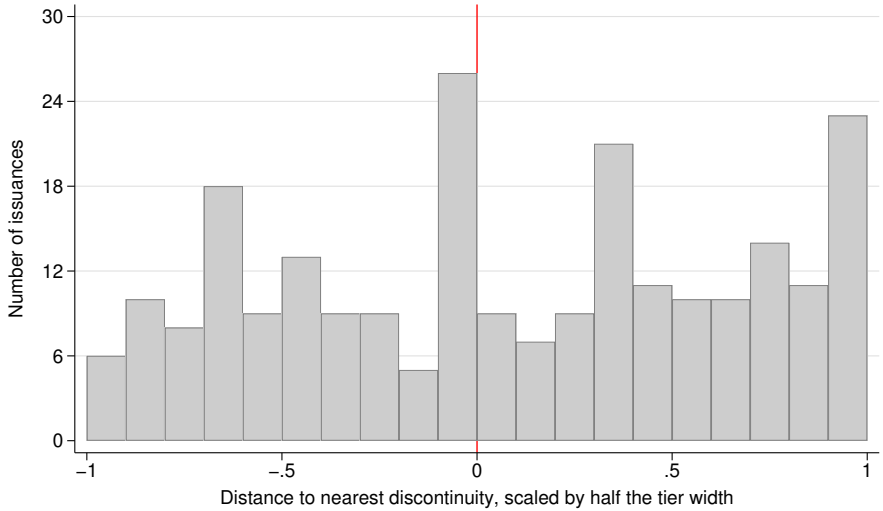
Note: This figure shows the predicted guarantee of the median director. The red line reflects the 1919–1925 rule, the blue line is the 1925–1930 rule, both backed out from the step function seen in figure B5. The open dots indicate observations where one of the bankers had an interlocked directorate, they are not used to infer the rule. Throughout our period of analysis the directors of the bank alternate and consist of either 3 or 4 persons (excluding president and secretary). The median director is based on 1) the one director that is present in the sample throughout the period (Jacob Bierens de Haan) and 2) is corrected for deviations by one of the bank directors. That is, if this bank director had a different personal guarantee than the other directors, we use the median value of the directors. Bank and banker guarantees are in thousands of currency of issuance. Includes only observations where the bank is one of the lead underwriters. Bank underwriting amount is displayed in levels.

Figure 2: Within bin comparison



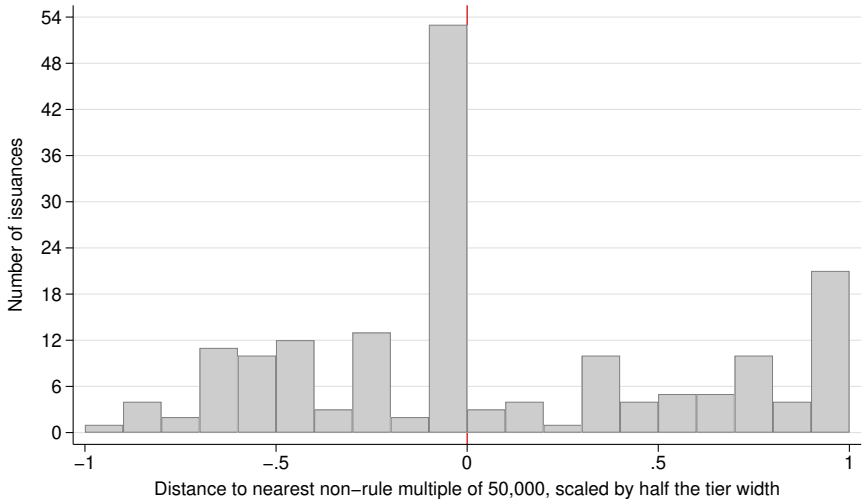
Note: This figure provides an example of some of the bins used in our regression analyses. We use bin fixed effects to compare observations within a bin, where the bank's guarantee is relatively similar while the banker's guarantee jumps around the discontinuity. The banker's guarantee displayed here is the median director's guarantee. The figure zooms in into 5 bins in the bank's guarantee range of 450.000 - 1.575.000. Each bin is defined from one middle point of a step until the next middle point. Vertical gray dashed lines reflect bin borders. The solid black line is the rule for 1925–1930. Bank and banker guarantee are thousands of the currency of issuance. Includes only observations where the bank is one of the lead underwriters.

Figure 3: Normalized distance of bank guarantees to the nearest discontinuity



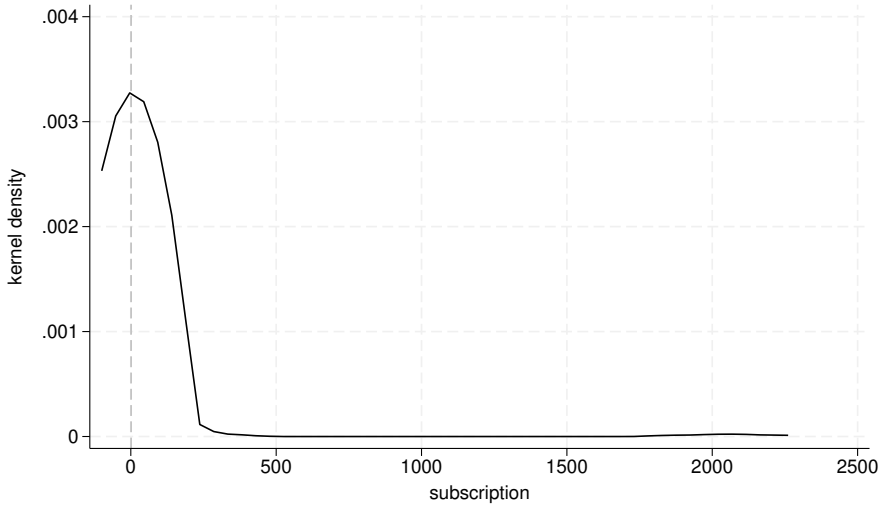
Note: This figure shows the distribution of the normalized distance of bank guarantees to the nearest discontinuity. Normalized distance = (guarantee - nearest discontinuity) / half the distance from it to the next discontinuity, on the side the guarantee falls. The measure is 0 when a guarantee sits exactly on a discontinuity (N=19) and +1 or -1 when it sits midway between two bins. The scaling is what allows the two regimes and the differently sized tiers to be pooled on one axis. Bins are closed on the right, so the bar left of the red line are the last bank underwriting amounts before the banker's stake jumps (N=26, of which N=19 is exactly on the discontinuity). Everything to the right of the red line is already above a discontinuity within a bin. Includes only observations where the bank is one of the lead underwriters, with one observation per issuance (N = 238 of 239; |distance| > 1 not shown). Bin width is 0.1, which reflects width between 2,500 and 127,500 in the currency of issuance (median = 10,000).

Figure 4: Normalized distance of bank guarantees to the nearest counterfactual discontinuity

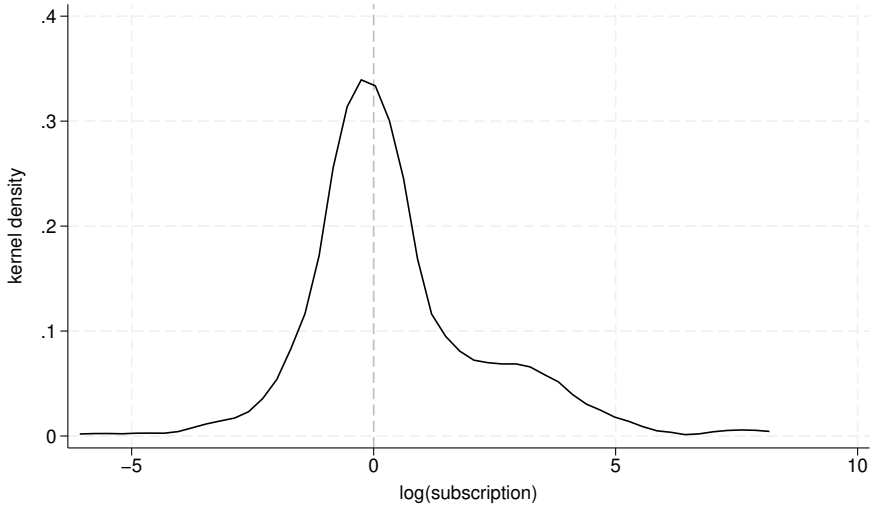


Note: This figure shows the distribution of the normalized distance of bank guarantees to the nearest round number that is not a discontinuity. Placebo thresholds are imposed on every 50,000 multiple, and thresholds that were part of the rule are dropped (61 out of 239 issuances). Each remaining guarantee is assigned to the placebo threshold nearest to it. Normalized distance = (guarantee - nearest discontinuity) / half the distance from it to the next discontinuity. The distance is 25,000 on both sides of the placebo threshold, since every 50,000 is considered a threshold. Bins are closed on the right, so the bar left of the red line are the last bank underwriting amounts before a placebo threshold (N= 53 issuances, of which 51 sitting exactly on a round number). Everything to the right of the red line is already above a placebo threshold within a bin. The bar at +1 holds 21 issuances, 20 of them exact multiples of 25,000 lying midway between two placebo thresholds, ties being assigned to the lower threshold. Includes only observations where the bank is one of the lead underwriters, with one observation per issuance (N = 178 of 239; no observation falls outside |distance| $\leq$ 1 by construction). Bin width is 0.1, which reflects a constant 2,500 guilders.

Figure 5: Distribution of subscription rate of issuance



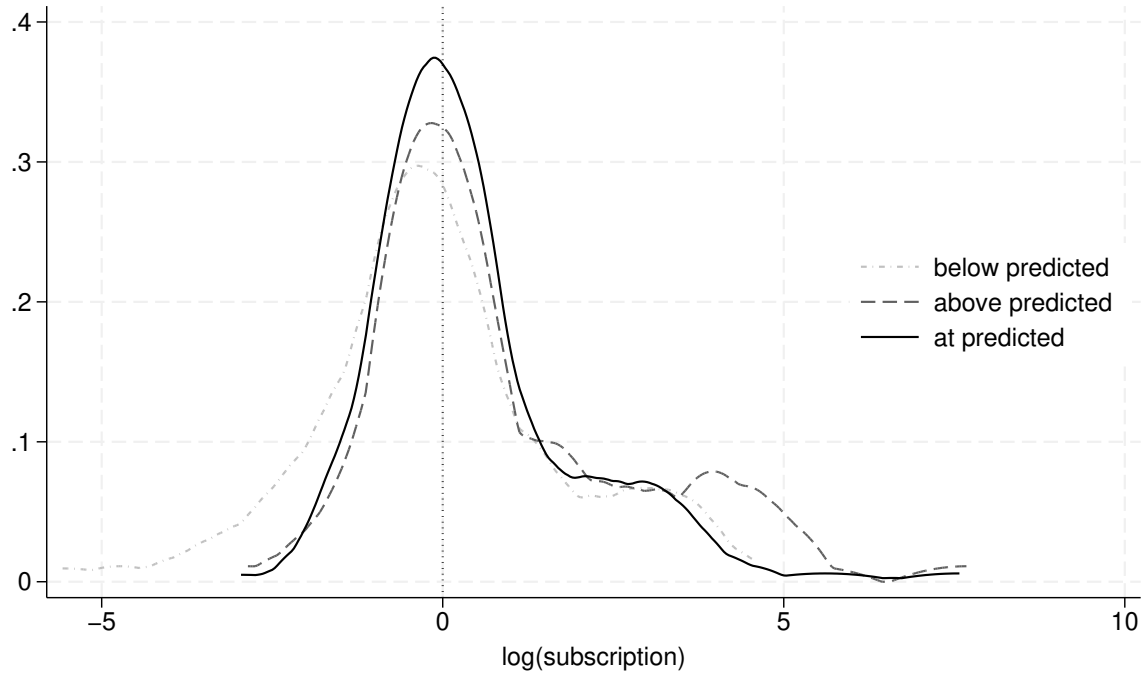
(a) Times subscribed to issuance



(b) Log(subscription rate) of issuance

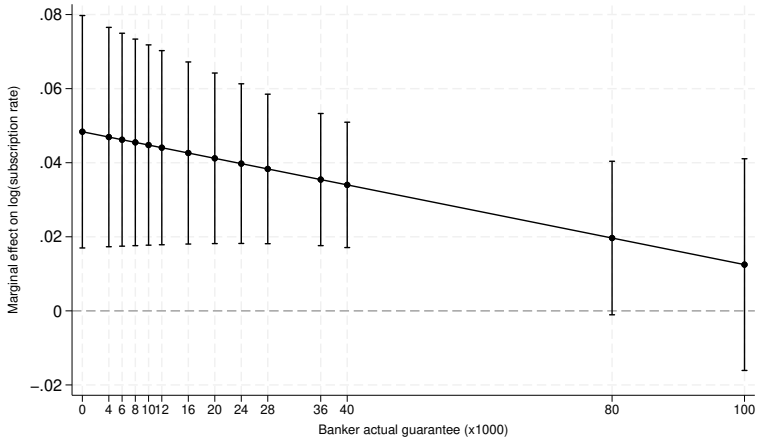
Note: The panels in this figure display the kernel density of our main dependent variable, the subscription rate. Panel (a) shows the actual subscription rate as ratio of the total securities outstanding at the security issuance. In this case, 1 means that the issue is fully subscribed. Panel (b) shows the log transformation of the subscription rate, where 0 means that the issue is fully subscribed. Both panels include all observations where the bank is one of the lead underwriters and the bank's guarantee does not exceed 5 million in the currency of issuance.

Figure 6: Banker guarantees and subscription to the issuance

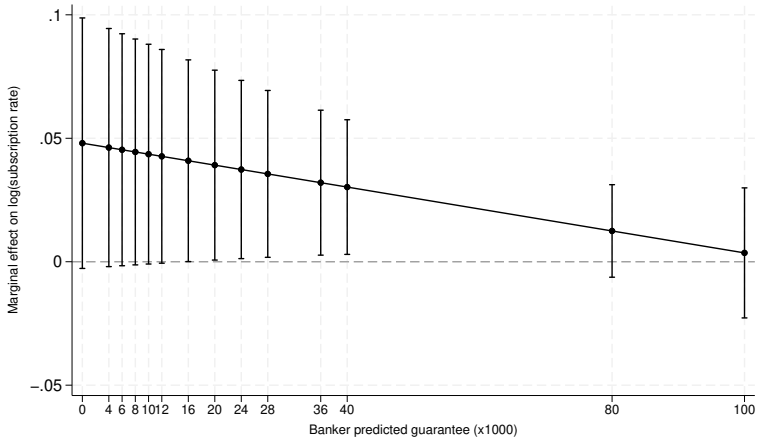


Note: This figure shows the kernel density of success of the issuance as log(subscription) for different sorts of bankers' guarantees. The solid line represents the issues where the bankers underwrite exactly the predicted amount. The dashed (dotted) line shows issues where the bankers' guarantees are higher (lower) than predicted. All mass right of the zero line is oversubscribed; all mass left of the zero is undersubscribed. Includes only observations where the bank is one of the lead underwriters and the bank's guarantee does not exceed 5 million in the currency of issuance.

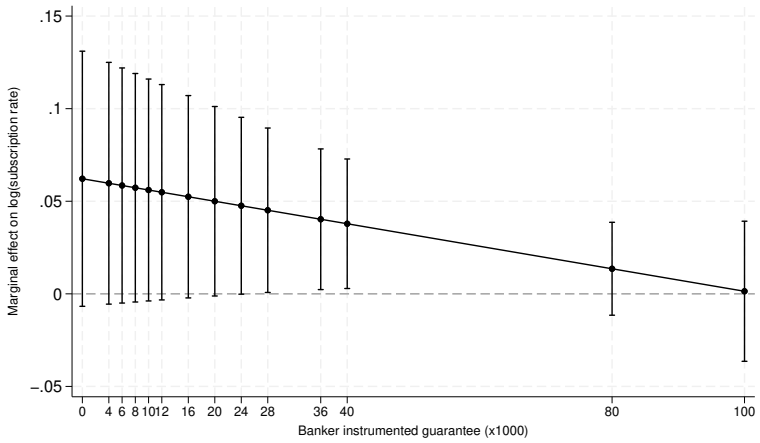
Figure 7: Marginal effects at each bankers' guarantee



(a) OLS



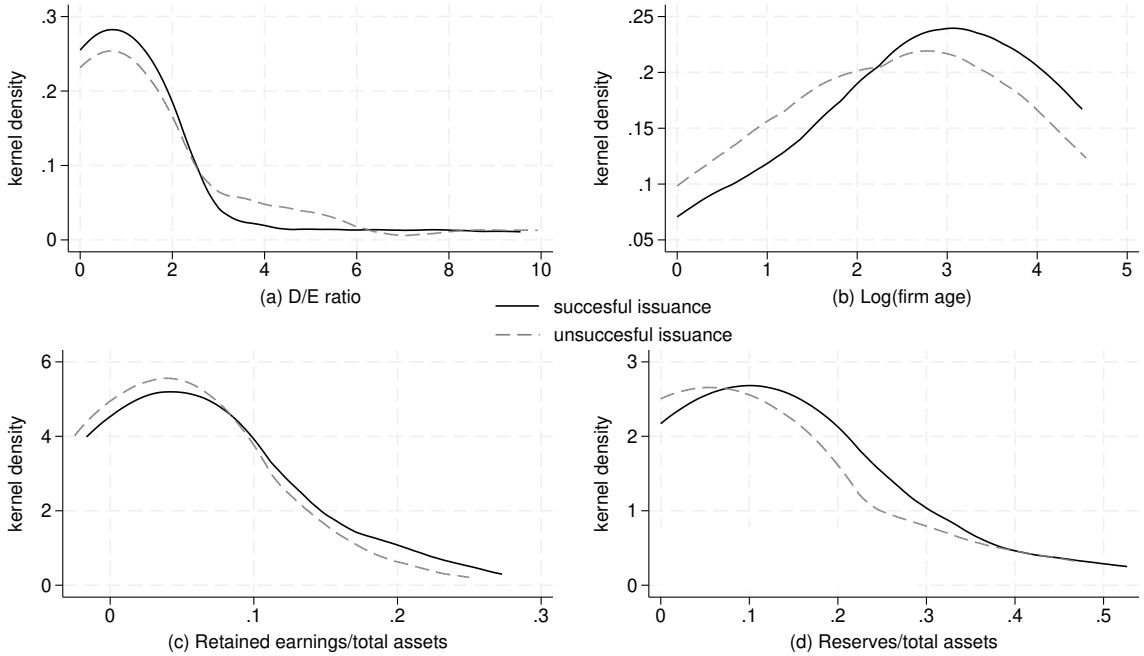
(b) RF



(c) IV

Note: The panels of this figure show the marginal effects at each point in the rule for banker guarantees. Panel (a) shows the OLS regression with an additional squared actual banker guarantee, panel (b) shows the reduced form regression with an additional squared predicted banker guarantee, and panel (c) displays the IV results with the instrumented banker guarantee and an additional instrumented squared banker guarantee. All panels include only observations where the bank is one of the lead underwriters and the bank's guarantee does not exceed 5 million in the currency of issuance.

Figure 8: Ex ante firm risk measures



Note: The panels of this figure show the kernel density of balance sheet variables for issuing firms, split in security issues that were successful (solid line) versus issues that were not fully subscribed (dashed line). We consider every issuance above 90% subscription as successful. Panel (a) shows total debt over total equity (= capital outstanding + retained earnings) (trimmed at 10 for visual representation) of the issuing firm at the year before issuance. Panel (b) shows the logarithm of the age of the firm at the time of issuance (where all firms from 0-12 months are coded as age 1). Panel (c) displays the retained earnings over total assets of the firm in the year before the security issuance. Panel (d) displays the reserves of the firm over total assets in the year before the issuance. All data are (balance sheet) information in the prospectus published before the security issuance, or if missing from 'Van Oss' Effectenboek' firm financial information book in the year of the issuance. All panels include only observations where the bank is one of the lead underwriters and the bank's guarantee does not exceed 5 million in the currency of issuance.

Tables

Table 1: Overview of yearly security issuances guaranteed by the bank

	Bank guarantees	N	Stocks (%)	IPO (%)	Regular secondary (%)	Rights issue (%)	Preferred issue (%)	Corp. bonds (%)	Govt. bonds (%)	Foreign (%)	Guilders (%)	Successful (%)
1919	14,157	17	65	0	0	41	24	0	35	0	100	71
1920	17,690	18	61	6	0	33	22	17	22	11	100	61
1921	9,925	11	9	0	0	9	0	82	9	0	100	64
1922	7,604	3	33	0	0	33	0	33	33	33	100	33
1923	6,720	5	0	0	0	0	0	60	40	20	100	60
1924	10,227	6	17	0	0	17	0	0	83	50	67	83
1925	23,549	17	0	0	0	0	0	35	65	59	59	82
1926	30,659	27	11	4	0	4	4	52	37	70	52	78
1927	46,192	38	8	5	3	0	0	42	50	82	39	55
1928	43,908	42	26	2	14	7	2	43	31	71	43	45
1929	20,731	21	29	0	19	10	0	38	33	81	33	43
1930	37,653	34	6	0	6	0	0	74	21	79	59	53

Note: This table provides an overview of the security issues guaranteed in our main regression sample. The first column, Bank guarantees, is the total amount guaranteed by the bank that year, in thousands of guilders; the second is the yearly number of unique bank guarantees. Every remaining column is a percentage of that year's issues. Stocks, corporate bonds and government bonds are mutually exclusive and sum to 100 percent. The four columns between Stocks and Corp. bonds break the equity issues down by how they were placed and sum to the Stocks column: IPOs, regular secondary offerings, the rights issues solely open to existing shareholders, and the preferred issues with preferential treatment to existing shareholders but open to all. Percentages are rounded to the nearest integer, so a displayed row may add up to one more or one less than these identities imply. The last three columns are the percentage of issues from foreign issuers, denominated in the domestic currency (guilders), and successful. The table only includes observations for which the bank is one of the lead underwriters and where the total bank guarantee is higher than 50 thousand but does not exceed 5 million in the currency of issuance.

Appendix

A Data construction securities returns

Our price panel covers the 239 securities underwritten by the NHM over a window of thirteen months. Every security's window starts in the month the security was listed, and records the low and the high quotation in each of those months, which results in 3,120 observations. Prices come from two printed sources. The primary source is the annual price yearbooks of the Amsterdam Stock Exchange, the *tabellenboeken*, which report a monthly low and high for every listed security. We digitized the complete volumes for 1921 through 1931 from page scans using a vision model, which yields 272,122 observations at the level of security, year and period. We matched our securities to the printed rows using name similarity together with agreement against price checkpoints that we collected by hand. The *tabellenboeken* volumes for 1919, 1920 and 1924 are summary volumes that report annual highs and lows only. For the 46 securities whose window falls in the resulting gaps, we read the last trading day of each month directly from the *Prijscourant*, the daily official price list of the Amsterdam Stock Exchange. Each security was located there by the printed section of the list, and its identity confirmed on coupon dates, denomination, loan year and coupon rate. Every observation flagged by our consistency checks, whether an implausible move from one month to the next, a low above the high, or a break against the printed annual figures, was read again by hand from the original scan.

Many securities only contain a low price for a month, or no price altogether: these are indications that the security was very infrequently traded. We compute a measure of not-traded months to proxy the illiquidity of a security. A month counts as not-traded when neither a high nor a low price is printed for it, and as half not-traded when only a low is printed.[38]

[38]The low and the high prices in the yearbooks are the minimum and the maximum over all trading days of the month, whereas a Prijscourant observation is a single day's quotation. Each Prijscourant value rests on two independent readings of the scan, one that locates the row on the full page and one that reads a close crop of it, and every value that reached the panel was checked by hand against that crop. For end of month daily prices that include only a low price, we also check the second to last day to slightly lower the risk of overestimating the number of not-traded months for the securities from the Prijscourant. One observation, Deli Spoorweg Maatschappij in March 1926, prints a low above its high in the source itself, and we adopt the prices as is from the source.

Table 2: Summary Statistics

	Mean	SD	P1	P50	P99	N
Bank guarantee (x1000)	1005.59	907.74	100.00	700.00	4175.00	1391
Actual banker guarantee (x1000)	20.52	21.70	0.00	15.00	100.00	1391
Predicted banker guarantee (x1000)	20.90	21.33	0.00	15.00	100.00	1391
Share of bank in syndicate	0.34	0.25	0.08	0.25	1.00	239
Log(subscription)	0.44	1.79	-3.18	-0.00	5.61	239
Times oversubscribed	25.58	189.08	0.04	1.00	273.42	239
Underwriting fee	3.04	1.81	0.38	2.88	8.50	228
Number of banks in syndicate	3.94	2.24	1.00	4.00	12.00	230
Successful issuance dummy	0.59	0.49	0	1	1	239
Fully subscribed	0.49	0.50	0	0	1	239
Public sector dummy	0.36	0.48	0	0	1	239
Foreign dummy	0.59	0.49	0	1	1	239
Guilders dummy	0.59	0.49	0	1	1	239
Interlocked directorate dummy	0.15	0.35	0	0	1	239
Equity dummy	0.21	0.41	0	0	1	239
IPO dummy	0.03	0.17	0	0	1	239
Main lender	0.17	0.37	0	0	1	239
First lead	0.71	0.45	0	1	1	239
Rights issue	0.09	0.29	0	0	1	239
Preferred issue	0.21	0.41	0	0	1	239
Issuance/size	0.13	0.19	0.00	0.07	0.94	166
Retained earnings/total assets	0.10	0.12	0.00	0.05	0.50	166
Profit/total assets	0.05	0.06	-0.02	0.03	0.25	166
Age of firm at issuance	24.83	25.41	1.00	17.00	94.00	151
Leverage (D/E)	2.12	4.95	0.00	0.69	23.54	141
Public budget surplus (f)/population	-0.32	13.41	-42.93	1.13	16.95	36
Outstanding public debt (f)/population	228.59	196.71	5.14	198.24	683.87	55

Note: This table reports summary statistics for the main regression sample where the bank is one of the lead underwriters and the total bank guarantee is at least 50 thousand and does not exceed 5 million in the currency of issuance. The first three variables are reported on the banker-issuance level. All other variables are reported at the issuance level. Bank guarantee is the net guarantee of the bank, after extending amounts to syndicate members or sub-participants (excluding the bank's bankers). Share (%) of bank in the syndicate is the bank's gross guarantee divided by the total syndicate guarantee. Number of banks in syndicate is the count of banks that are syndicate members. Log(subscription) is the logarithm of the number of securities subscribed by market participants over the total of issued securities. Times oversubscribed is the number of times an issuance has been subscribed to. Successful issuance dummy is equal to 1 if the subscription rate was above 90%. Fully subscribed is a dummy that equals one if the subscription rate is 100% or higher. Dummies equal 1 (0) if public (private) sector security, foreign (domestic), issued in domestic (foreign) currency, one of the bankers has an interlocked directorate, dummy for stocks (bonds), dummy for IPOs, dummy if the bank is the main lender of the firm, if the bank is the first lead of the syndicate and if the issuance is a rights issue or a preferred issue (which covers all preferential treatments for existing shareholders, including rights issues). Firm level variables have fewer observations, since the sample is also made up of (local) governments. Issuance/size is the total security issuance amount of the firm over its total assets measured in the year before. Underwriting fee (%) is the commission that the bank(ers) received over their underwriting amount. Retained earnings, profit and leverage are all weighted by total assets in the year before issuance. Leverage (D/E) is total debt weighted by total equity. The last two rows shows summary statistics for issuing (local) governments. Both outstanding debt and budget surplus(deficit) are in guilders and are weighted by the population in the year before the issuance.

Table 3: Balancing table

	(1) Below discontinuity Mean/(SE)	(2) Above discontinuity Mean/(SE)	(1)-(2) Pairwise t-test Mean difference
Bank guarantee (x1000)	962.212	1045.218	-83.007***
	(89.092)	(78.106)	
Actual banker guarantee (x1000)	17.854	22.962	-5.108***
	(1.849)	(1.936)	
Predicted banker guarantee (x1000)	17.524	23.983	-6.459***
	(1.730)	(1.988)	
Log(subscription)	0.205	0.682	-0.476*
	(0.128)	(0.186)	
Fully subscribed	0.434	0.547	-0.114
	(0.047)	(0.045)	
Share of bank in syndicate	0.354	0.338	0.015
	(0.023)	(0.023)	
Number of banks in syndicate	3.785	4.038	-0.253
	(0.199)	(0.213)	
Underwriting fee	3.045	3.040	0.005
	(0.168)	(0.174)	
Public sector dummy	0.361	0.362	-0.000
	(0.045)	(0.043)	
Foreign dummy	0.568	0.602	-0.035
	(0.047)	(0.044)	
Guilders dummy	0.577	0.608	-0.031
	(0.047)	(0.044)	
Interlocked directorate dummy	0.130	0.164	-0.034
	(0.031)	(0.034)	
Equity dummy	0.232	0.195	0.037
	(0.040)	(0.036)	
IPO dummy	0.018	0.041	-0.023
	(0.013)	(0.018)	
Main lender	0.178	0.162	0.015
	(0.036)	(0.033)	
First lead	0.694	0.730	-0.036
	(0.043)	(0.040)	
Rights issue	0.098	0.089	0.008
	(0.028)	(0.026)	
Number of bankers	5.102	5.457	-0.354
	(0.187)	(0.134)	
Observations	664	727	1391
Clusters	114	125	239

Note: This balancing table shows the means of the observations of our main regression sample below and above the discontinuities, collapsed to one level. Column 1 (2) reports the mean and standard errors of observations below (above) each discontinuity. The pairwise t-test, displayed in the last column, includes bin fixed effects. Observations are clustered by issuance. Log(subscription) is the logarithm of number of securities subscribed to over the total outstanding in the issuance. Bank guarantee is the net guarantee of the bank, after extending amounts to syndicate members or sub-participants (excluding the bank's bankers). Log(subscription) is the logarithm of the number of securities subscribed by market participants over the total of issued securities. If the issuance was fully subscribed, the log(subscription) is zero. Full subscription dummy is equal to 1 if the subscription rate was equal to or above 100%. Share (%) of bank in the syndicate is the bank's gross guarantee divided by the total syndicate guarantee. Number of banks in syndicate is the count of banks that are syndicate members. Underwriting fee (%) is the commission that the bank(ers) received over their underwriting amount (only available for 110 below and 118 securities above the threshold). Dummies equal 1 (0) if public (private) sector security, foreign (domestic), issued in domestic (foreign) currency, one of the bankers has an interlocked directorate, dummy for stocks (bonds), dummy for IPOs, dummy if the bank is the main lender of the firm, a dummy if the bank is the first lead of the syndicate and a dummy for rights issues (solely open to existing shareholders). The number of bankers reflects the number of bankers personally underwriting in each issuance. This number varies over time due to the different number of bankers in the sample. It is included here to show that the number of bankers in the firm varied between 5 and 6. The table only includes observations for which the bank is one of the lead underwriters and where the total bank guarantee is higher than 50 thousand but does not exceed 5 million in the currency of issuance.

Table 4: Bankers' guarantees and success of issuance, with fixed effects

	(1)	(2)	(3)	(4)	(5)
		(Co)-lead in syndicate			First lead
			OLS		
Actual banker guarantee (x1000)	0.027***	0.026***	0.029***	0.032***	0.032***
	(0.007)	(0.007)	(0.008)	(0.008)	(0.009)
R^2	0.180	0.204	0.207	0.232	0.278
			RF		
Predicted banker guarantee (x1000)	0.015**	0.015**	0.019**	0.020**	0.030**
	(0.007)	(0.007)	(0.009)	(0.010)	(0.013)
R^2	0.150	0.176	0.178	0.198	0.256
			IV 2nd stage		
Actual banker guarantee (x1000)	0.017**	0.017**	0.023**	0.025**	0.040**
	(0.008)	(0.008)	(0.011)	(0.012)	(0.016)
R^2	0.035	0.032	0.039	0.069	0.085
			IV 1st stage		
			Actual banker guarantee		
Predicted banker guarantee (x1000)	0.872***	0.881***	0.820***	0.807***	0.758***
	(0.092)	(0.086)	(0.108)	(0.107)	(0.126)
R^2	0.259	0.276	0.206	0.243	0.271
IV F-stat	89.76	105.52	57.28	56.76	36.20
Observations	1,391	1,391	1,391	1,391	992
Clusters	239	239	239	239	170
Mean of dep. var.			0.454		0.453
SD of Actual banker guarantee (x1000)			21.701		19.643
Bin FE	Y	Y	Y	Y	Y
Year FE	Y	Y	Y	Y	Y
Industry FE	N	Y	Y	Y	Y
Director FE	N	N	Y	Y	Y
Controls	N	N	N	Y	Y

Note: This table presents estimates of actual and predicted banker guarantees for log(subscription to the issuance) as a measure of quality of the securities underwritten by the bank. We only include the observations for which the bank is one of the lead underwriters and where the total bank guarantee does not exceed 5 million in the currency of issuance. Bank guarantees below 50 thousand are excluded from the analysis. Log(subscription) is the logarithm of the subscription to the security issuance, which are the securities sold at issuance over the total securities offered. The explanatory variables are the actual or predicted banker guarantee (in thousands of issued currency). The rows present the results for OLS, reduced form, and IV regression including the first stage. All specifications have bin and year fixed effects. Column (2) includes industry fixed effects, column (3) director fixed effects. Column (4) includes all controls and fixed effects, while (5) is the same specification as (4) but contains only observations where the bank is the first lead of the syndicate, i.e. appears first on the prospectus. Controls included in (4) and (5) are: bank share in syndicate, a dummy for public issuers, stocks, foreign, interlocked directorate, a main bank dummy, an IPO dummy, and a rights-issue dummy. Standard errors (in parentheses) are clustered at the issuance-level. The R-squared reported in the first-stage panel is the within R-squared after absorbing the fixed effects, and the R-squared in the IV second stage is computed after partialling out the fixed effects; R-squared values are therefore not directly comparable across panels. ***, ** and * indicate significance at 1, 5 and 10% level, respectively.

Table 5: Banker guarantees and deal size

	(1) Log(subscription) Base	(2) Log(subscription) Interacted	(3) Log(subscription) $\leq$ 2 mln
		OLS	
Actual banker guarantee (x1000)	0.032***	0.025**	0.023**
	(0.008)	(0.011)	(0.011)
× above 2 mln		0.013	
		(0.014)	
R^2	0.232	0.234	0.232
		RF	
Predicted banker guarantee (x1000)	0.020**	0.060***	0.065***
	(0.010)	(0.022)	(0.023)
× above 2 mln		−0.050**	
		(0.022)	
R^2	0.198	0.205	0.239
		IV 2nd stage	
Actual banker guarantee (x1000)	0.025**	0.084**	0.086**
	(0.012)	(0.037)	(0.037)
× above 2 mln		−0.068**	
		(0.034)	
R^2	0.069	−0.010	−0.041
		IV 1st stage	
	(1)	(2a) (2b)	(3)
Dependent variable	Actual	Actual Actual × above 2 mln	Actual
Predicted banker guarantee (x1000)	0.807***	0.720*** 0.013	0.746***
	(0.107)	(0.171) (0.079)	(0.156)
× above 2 mln		0.097 0.857***	
		(0.188) (0.125)	
R^2	0.243	0.253 0.418	0.215
Observations	1,391	1,391	1,200
Clusters	239	239	206
Kleibergen-Paap F (joint in (2))	56.76	10.57	22.99
Sanderson-Windmeijer F		21.16 71.71	
Bin FE	Y	Y	Y
Year FE	Y	Y	Y
Industry FE	Y	Y	Y
Director FE	Y	Y	Y
Controls	Y	Y	Y

Note: This table tests whether the effect of banker guarantees on log(subscription) differs between issuances above and at or below 2 million of net bank guarantee. Column (1) repeats the baseline specification of Table 4, column (4). Column (2) adds an interaction between the banker guarantee and an indicator for issuances above 2 million; in the IV columns both the guarantee and the interaction are instrumented, by the predicted guarantee and its interaction, and the indicator itself enters as an exogenous control. Column (2) therefore has two first-stage regressions, one per endogenous regressor, each on both excluded instruments: (2a) for the guarantee and (2b) for its interaction. Column (3) restricts the sample to issuances at or below 2 million. Because column (3) is a subsample of columns (1) and (2), the columns are not independent and their standard errors should not be compared directly; the interaction in column (2) is the test of whether the two slopes differ. All specifications include the same set of controls (bank share in syndicate, a dummy for public issuers, stocks, foreign, interlocked directorate, a main bank dummy, an IPO dummy, and a rights-issue dummy) and have bin, year, industry and director fixed effects. Standard errors (in parentheses) are clustered by issuance. The Kleibergen-Paap statistic is the rk Wald F; in column (2) it tests joint identification of both endogenous regressors, and the Sanderson-Windmeijer F, reported under each first stage, tests identification of each regressor conditional on the other. The R-squared reported in the first-stage panel is the within R-squared after absorbing the fixed effects, and the R-squared in the IV second stage is computed after partialling out the fixed effects; R-squared values are therefore not directly comparable across panels. ***, ** and * indicate significance at 1, 5 and 10% level, respectively.

Table 6: Subscription to issuance for each type of director guarantee

	(1) Log(subscription) President	(2) Log(subscription) Director	(3) Log(subscription) Secretary
		OLS	
Actual banker guarantee (x1000)	0.026***	0.041***	0.058***
	(0.008)	(0.011)	(0.014)
R^2	0.234	0.248	0.252
		RF	
Predicted banker guarantee (x1000)	0.031*	0.039*	0.052*
	(0.016)	(0.020)	(0.027)
R^2	0.205	0.205	0.205
		IV 2nd stage	
Actual banker guarantee (x1000)	0.041*	0.048*	0.067*
	(0.024)	(0.026)	(0.037)
R^2	0.056	0.086	0.092
		IV 1st stage Actual banker guarantee	
Predicted banker guarantee (x1000)	0.754***	0.805***	0.775***
	(0.215)	(0.202)	(0.205)
R^2	0.162	0.183	0.174
IV F-stat	12.31	15.89	14.27
Observations	237	237	237
Mean of dep. var.	0.444	0.444	0.444
SD of Actual banker guarantee (x1000)	27.104	20.524	15.022
Bin FE	Y	Y	Y
Year FE	Y	Y	Y
Industry FE	Y	Y	Y
Controls	Y	Y	Y

Note: This table presents estimates of president, director and secretary underwriting that are part of the bank's executive board throughout the period of analysis. The dependent variable is the logarithm of the subscription to the security issuance. We only include the observations for which the bank is one of the lead underwriters and where the total bank guarantee does not exceed 5 million in the currency of issuance. The rows present the results for OLS, RF, and IV regression including the first stage. All columns include the following set of controls: bank share in syndicate, a dummy for public issuers, stocks, foreign, interlocked directorate, a main bank dummy, an IPO dummy, and a rights-issue dummy. All columns include bin, year and industry fixed effects. Robust standard errors are in parentheses. The R-squared reported in the first-stage panel is the within R-squared after absorbing the fixed effects, and the R-squared in the IV second stage is computed after partialling out the fixed effects; R-squared values are therefore not directly comparable across panels. ***, ** and * indicate significance at 1, 5 and 10% level, respectively.

Table 7: Bankers' guarantees and firm risk

	(1) Log(subscription)	(2) PC 1	(3) PC 2
		OLS	
Actual banker guarantee (x1000)	0.044***	0.011	0.017*
	(0.016)	(0.010)	(0.009)
R^2	0.314	0.607	0.316
		RF	
Predicted banker guarantee (x1000)	0.028**	0.022*	0.014
	(0.014)	(0.013)	(0.013)
R^2	0.281	0.609	0.306
		IV 2nd stage	
Actual banker guarantee (x1000)	0.035**	0.027*	0.017
	(0.017)	(0.014)	(0.015)
R^2	0.073	0.209	0.183
		IV 1st stage	
Predicted banker guarantee (x1000)		0.817***	
		(0.099)	
R^2		0.310	
IV F-stat		68.48	
Observations	793	793	793
Clusters	137	137	137
Mean of dep. var.	0.452		
SD of Actual banker guarantee (x1000)	21.212	21.212	21.212
Bin FE	Y	Y	Y
Year FE	Y	Y	Y
Director FE	Y	Y	Y
Industry FE	Y	Y	Y
Controls	Y	Y	Y

Note: This table presents estimates of actual and predicted banker underwriting for firm ex ante risk. We only include the observations for which the bank is the lead underwriter and where the total bank guarantee does not exceed 5 million in the currency of issuance. Log(subscription) is the logarithm of the subscription to the security issuance, which are the securities sold on the stock market over the total securities offered. PC1 and PC2 are the first two principal components extracted from a set of firm balance sheet variables (see Table C1 for the loadings). PC1 loads primarily on financial health (leverage, short term debt over total debt, profit over total assets), while PC2 loads primarily on firm maturity (age, retained earnings over total assets, issuance size over firm size). All specifications include the same set of controls (bank share in syndicate, stocks, foreign, interlocked directorate, a main bank dummy, an IPO dummy, and a rights-issue dummy). All columns have fixed effects: bin, year, industry and director. Standard errors (in parentheses) are clustered by issuance. The R-squared reported in the first-stage panel is the within R-squared after absorbing the fixed effects, and the R-squared in the IV second stage is computed after partialling out the fixed effects; R-squared values are therefore not directly comparable across panels. ***, ** and * indicate significance at 1, 5 and 10% level, respectively.

Table 8: Buy-and-hold abnormal returns and banker guarantees

	(1)	(2)	(3)	(4)	(5)	(6)
	Months illiquid	Buy-and-hold abnormal returns				
		Unadjusted	P5	P10	P25	Balanced
Horizon		1m	1m	1m	1m	1m
			OLS			
Actual banker guarantee (x1000)	−0.021*	0.059*	0.080**	0.076**	0.074**	0.002
	(0.011)	(0.034)	(0.035)	(0.034)	(0.034)	(0.032)
R^2	0.415	0.375	0.374	0.373	0.379	0.575
			RF			
Predicted banker guarantee (x1000)	−0.052**	0.064	0.137**	0.126**	0.100**	0.089**
	(0.022)	(0.044)	(0.058)	(0.055)	(0.048)	(0.039)
R^2	0.423	0.373	0.374	0.373	0.377	0.579
			IV 2nd stage			
Actual banker guarantee (x1000)	−0.059**	0.072	0.153**	0.141**	0.112**	0.092**
	(0.026)	(0.049)	(0.067)	(0.064)	(0.055)	(0.041)
R^2	0.056	0.251	0.250	0.249	0.252	0.348
			IV 1st stage			
Predicted banker guarantee (x1000)	0.893***	0.898***		0.893***		0.968***
	(0.103)	(0.101)		(0.103)		(0.107)
IV F-stat	75.87	78.39		75.87		81.17
R^2 FS	0.315	0.320		0.315		0.415
Observations	1,322	1,264	1,322	1,322	1,322	846
Clusters	227	217	227	227	227	145
Mean of dep. var.	1.805	3.064	2.467	2.588	2.781	3.322
SD of Actual banker guarantee (x1000)	21.713	21.771	21.713	21.713	21.713	23.741
Bin FE	Y	Y	Y	Y	Y	Y
Year FE	Y	Y	Y	Y	Y	Y
Industry FE	Y	Y	Y	Y	Y	Y
Director FE	Y	Y	Y	Y	Y	Y
Controls	Y	Y	Y	Y	Y	Y

Note: This table presents regressions of ex post quality of issuances. Column (1) shows estimates for the number of months, over the 12 months after the issuance month, where a security is not traded: 1 month if no high or low price is reported, and 0.5 if only a low price is reported over the respective month. Columns (2) to (6) show buy-and-hold abnormal returns for underwritten securities, calculated over a horizon of 1 month after issuance. BHAR is computed as the cumulative simple return of the security minus the cumulative return of the market index over the same horizon multiplied by 100. Column (2) uses unadjusted returns without imputation. In columns (3) to (5), if a security is not traded in a month, we impute the missing value with the 5th, 10th or 25th percentile value of that stock or bond within a 3 year period window. Column (6) repeats the unadjusted specification on the balanced sample of securities with a reported price in all twelve months after the issuance month, where no value is imputed. All specifications include the same set of controls (bank share in syndicate, a dummy for public issuers, stocks, foreign, interlocked directorate, a main bank dummy, an IPO dummy, and a rights-issue dummy). All specifications have bin, year, industry and director fixed effects. Standard errors (in parentheses) are clustered by issuance. The R-squared reported in the first-stage panel is the within R-squared after absorbing the fixed effects, and the R-squared in the IV second stage is computed after partialling out the fixed effects; R-squared values are therefore not directly comparable across panels. ***, ** and * indicate significance at 1, 5 and 10% level, respectively.

Benchmarks

Abnormal returns are measured against two benchmarks, one for bonds and one for stocks, each constructed to reflect the alternative an investor in that market actually faced.

Bonds

We benchmark each bond against the Dutch government bond "on the run" at the time of issuance. That is, we use the most recently issued sovereign loan then trading, and take the midpoint of its monthly low and high. Contemporaries used the same reference point: underwriting agreements in the early part of our period routinely gave the syndicate the right to withdraw if a named government loan fell below a stated price, such as the 5% Nederland below 94.5% for the municipality of The Hague in February 1919, or the 5% Landsleening 1918 below 90% for the municipality of Amsterdam in May 1919.[39] Each security is matched to the loan on the run at its issuance date; we always use the *f.* 1,000 denomination, and where that loan is delisted before the end of a security's return window we fall back to the most recently issued government loan still quoted. For 1919, 1920 and 1924 the government series is read end of month from the *Prijscourant*.

Stocks

We benchmark each stock against a price index we construct from the weekly issues of the *Economisch-Statistische Berichten* (ESB) for 1919–1932. The way ESB reports prices changes twice: it prints tables by category until 1924, quotes prices within the text of the market report between 1924 and 1930, and from 1930 reproduces the index of De Bank voor Handel en Scheepvaart in Rotterdam. For every weekly issue we take the first quoted price of each security, which is the price least often affected by transcription error and is never missing. Weekly returns are compounded within the month and weighted by the previous month's market capitalisation, renormalised each month, and the index is chained from a base of one hundred. Securities quoted in dollars enter at a tenth of their nominal capital, reflecting the share held by Dutch investors (Veenendaal 1996). Returns spanning more than three weeks are set to missing, which is what absorbs the break in ESB's price overviews between 3 July 1929 and 8 January 1930. The index covers 71 securities in the first period and 177 in the second, spanning banking, industry, trade, mining, oil, rubber, tobacco, sugar, tea, shipping and, until 1930, the American market.

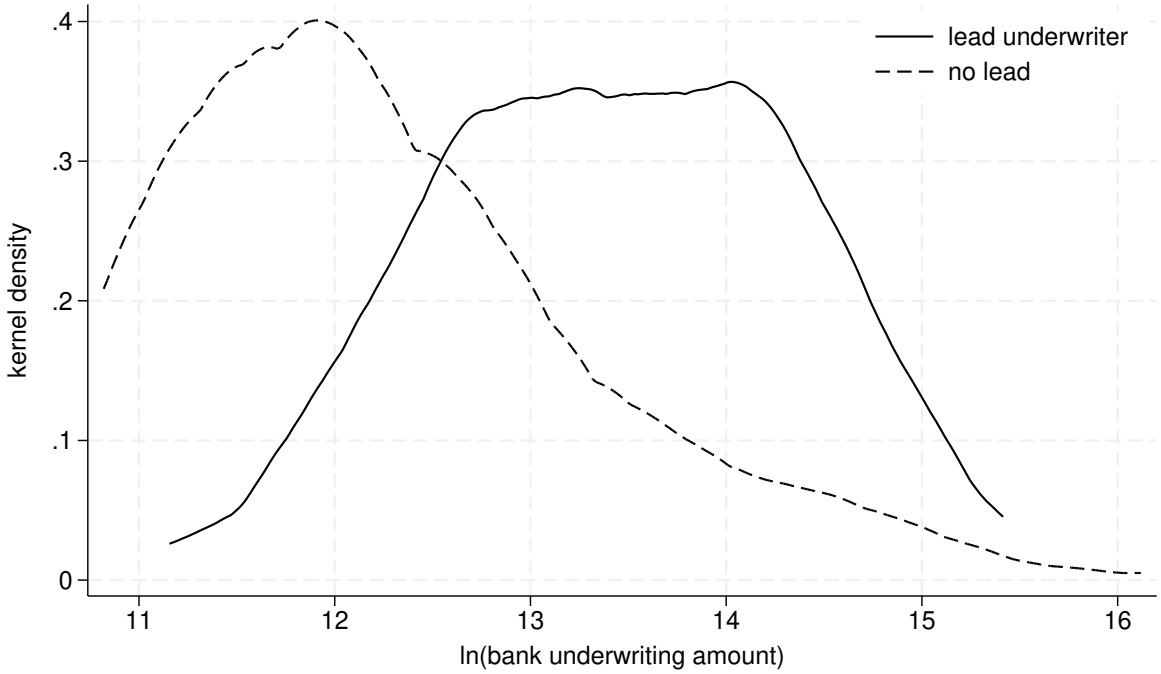
[39] Comparable clauses appear until 1924, after which the minutes replace them with general provisions allowing the syndicate to withdraw if conditions in the money or investment market make a successful placement unlikely. NL-HaNA, NHM, 442: 13-02-1919, 03-05-1919, 02-06-1919; 445: 15-09-1922.

Dividends

Neither benchmark is adjusted for dividends or coupons. Bonds are not adjusted for coupons in the underlying data, since prices were quoted without coupon payments, only to be calculated as an additional charge at the moment of the transaction (Brouwer, 1969). Stocks are not adjusted either. The exact payout date for dividend is difficult to identify, and since both the benchmark and the individual securities are not adjusted for dividend payouts, the comparison is still consistent. What remains is the difference between a security's own payout over the window and that of the index.

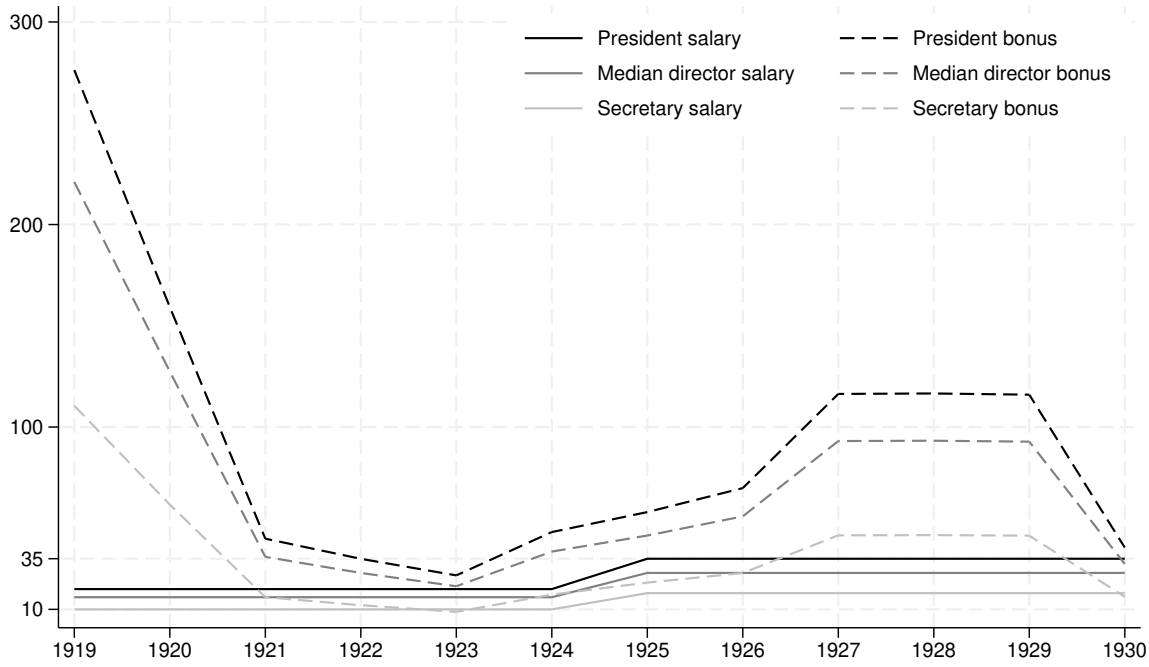
B Issuance characteristics

Figure B1: Distribution of bank's guarantee, by lead underwriter



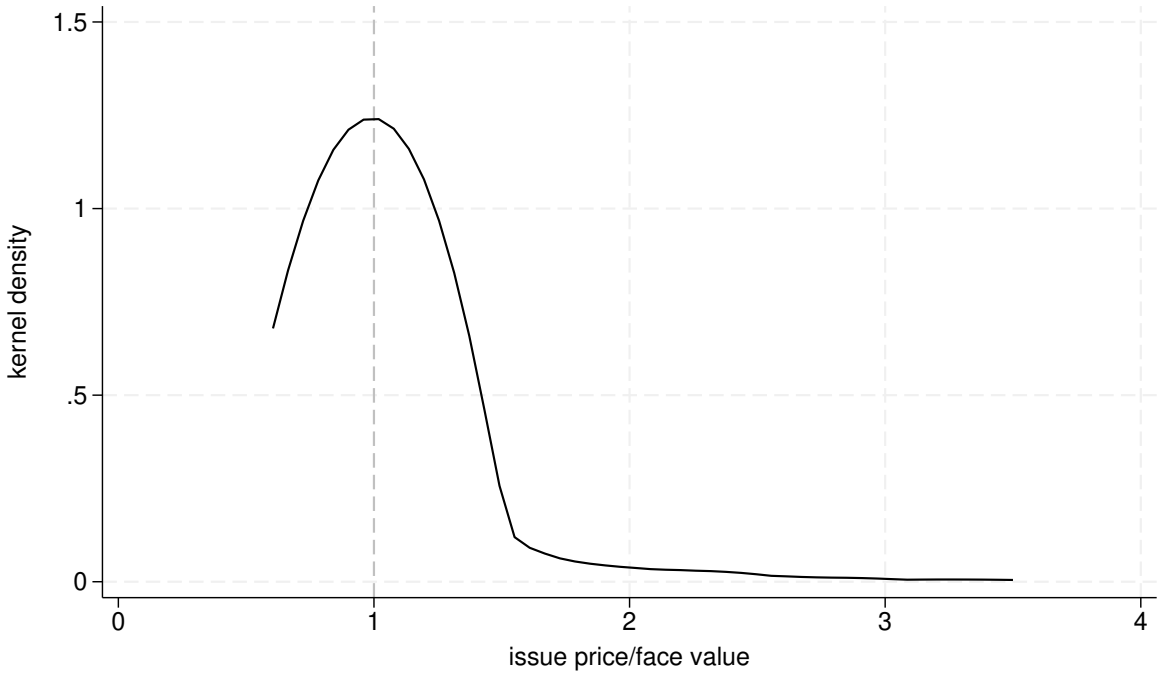
Note: This figure shows the kernel density of the natural logarithm of bank's guarantee in security issues. The solid line reflects the observations where the bank is one of the lead underwriters, while the dashed line represents the observations for which the bank is a participant in the syndicate. The densities include all observations below 5 million net bank's guarantee in the currency of issuance.

Figure B2: Banker compensation 1919-1930



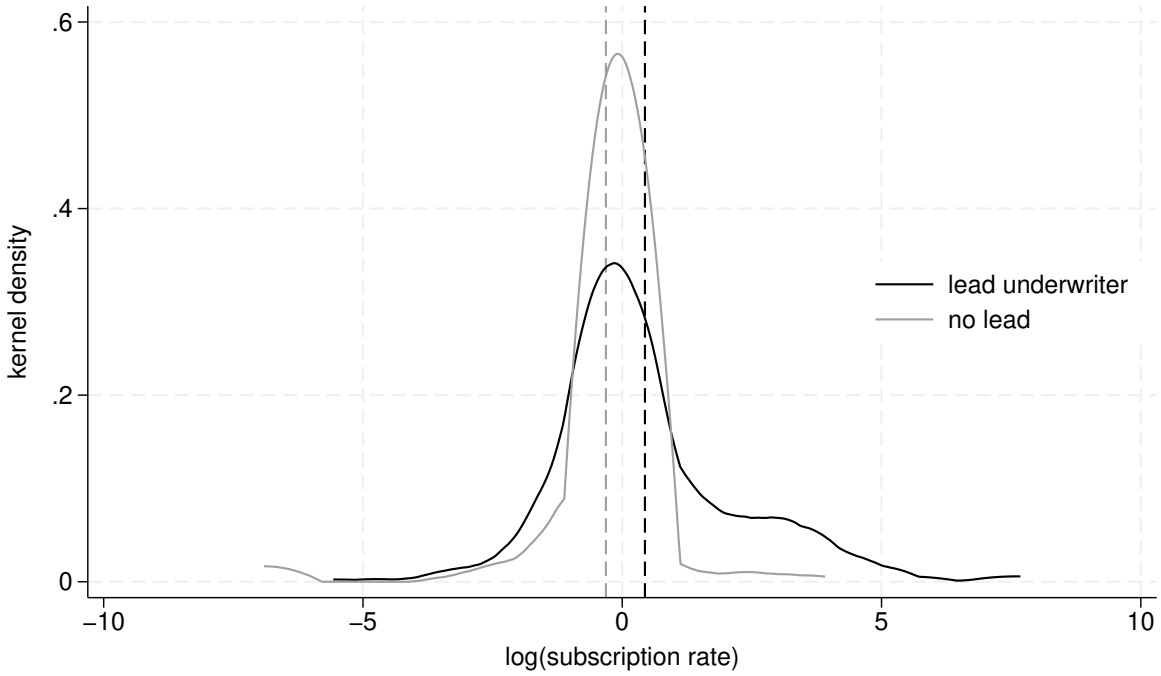
Note: This figure displays the annual salaries (solid lines) and tantièmes (dashed lines) for the president (black), median director (mid gray) and secretary (light gray). The base salaries range from $f.10.000$ to $f.35.000$ guilders in our sample period. The annual salary for the president is $f.20.000$ from 1919-1924, $f.35.000$ thereafter. For the directors, $f.16.000$ from 1919-1924, $f.$ 28.000 thereafter. For the secretary: $f.10.000$ from 1919-1924, $f.18.000$ thereafter. Directors enter and exit throughout the years, leading to pro rata salaries. The figure shows the 'median' director to display the compensation over a full year of a bank director. Amounts in thousands of guilders.

Figure B3: Issue price as a ratio of face value



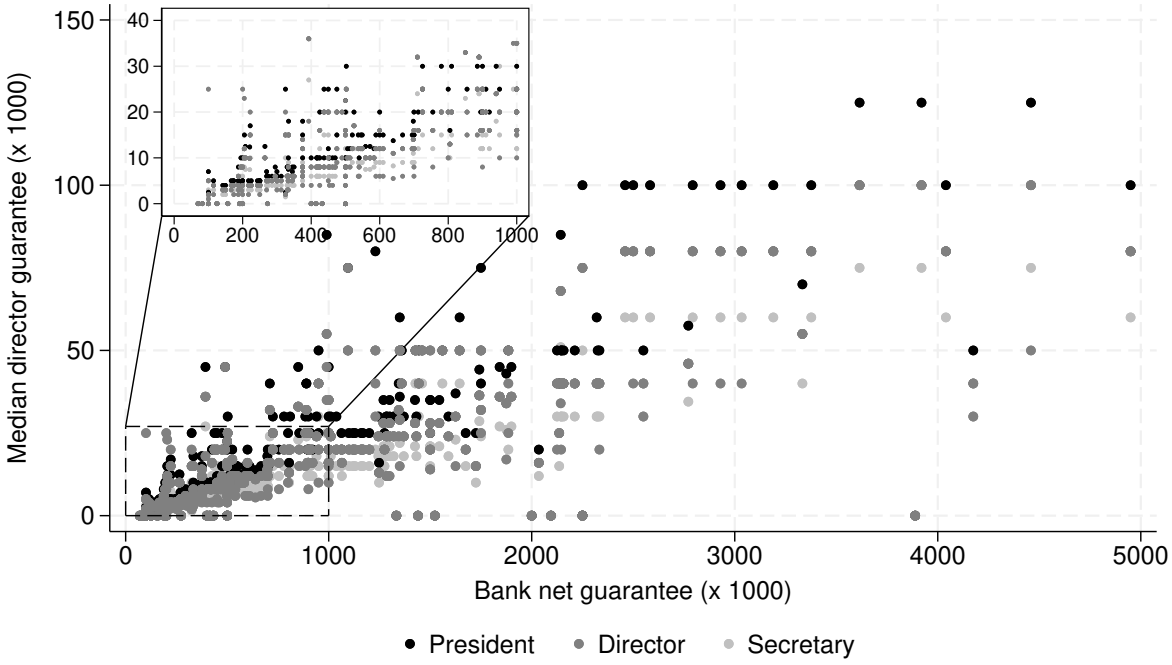
Note: This figure shows the kernel density of the issue price of the security issuance over the face value. This is the ratio of the price at which the securities are bought by the bank over the price at which the securities are sold to the market. Most issue prices are around par. Includes all observations where the bank is one of the lead underwriters and the bank's guarantee does not exceed 5 million in the currency of issuance.

Figure B4: Success of the issuance for participating or leading a syndicate



Note: This figure shows the distribution (kernel density) of the success of the security issues by leading or participating in a syndicate. Log(subscription) is the logarithm of the subscription to the security issuance, which are the securities sold on the stock market over the total securities offered. The black (dashed) line reflects the observations (mean) of bank lead underwriting issues, while the gray (dashed) line reflects observations (mean) of bank participations in other syndicates. Includes all observations where the bank is one of the lead underwriters and the bank's guarantee does not exceed 5 million in the currency of issuance.

Figure B5: Bank and bankers' guarantees



Note: This figure shows all the observations used in our regression sample. It shows guarantees for the president (black), median director (mid gray) and secretary (light gray). The bank and banker guarantees are in thousands of guilders. Includes all observations where the bank is one of the lead underwriters and the bank's guarantee does not exceed 5 million in the currency of issuance.

C PC analysis

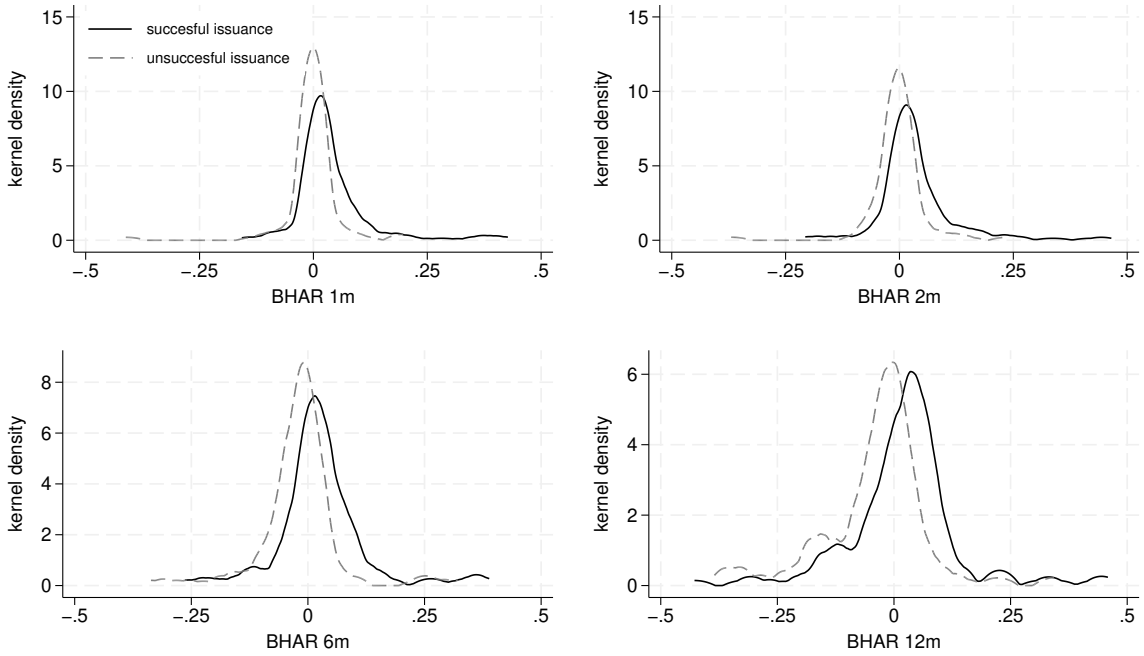
Table C1: Principal component analysis

	PC1	PC2	PC3	PC4	PC5	PC6
Leverage (Debt/Equity)	-0.548	-0.197	0.023	0.460	0.216	0.634
Retained earnings / Total assets	0.297	0.530	0.540	0.111	-0.361	0.445
Profit / Total assets	0.490	-0.072	-0.717	0.185	-0.251	0.379
Log (age of company)	-0.117	0.607	-0.240	0.618	0.136	-0.399
Issuance / Total assets	0.227	-0.550	0.320	0.587	-0.331	-0.299
ST debt/ Total debt	0.553	-0.062	0.187	0.122	0.795	0.093
Eigenvalue	1.658	1.378	0.848	0.813	0.698	0.605
Proportion	0.276	0.230	0.141	0.136	0.116	0.101
Cumulative	0.276	0.506	0.647	0.783	0.899	1.000

Note: This table reports the results of a principal component analysis of six firm balance sheet variables: leverage (debt/equity), retained earnings/total assets, profit/total assets, firm age at moment of issuance (in logs), the issuance ratio (issuance size relative to firm size) and short term debt over total debt. Each variable is standardized to zero mean and unit variance before the analysis. The sample consists of all observations in the corporate subsample (non-government securities for which the bank is a lead underwriter, the total bank guarantee does not exceed 5 million in the currency of issuance, and complete balance sheet data are available). The upper panel reports the loadings on the principal components. The lower panel reports the eigenvalues, the proportion of total variance explained by each component, and the cumulative proportion explained. The first two components have eigenvalues above one (Kaiser criterion) and together explain approximately 51% of the variation in the underlying variables. This normalization does not affect variance explained or component scores up to sign. Component scores constructed from PC1 and PC2 are used as dependent variables in Table 7 of the main text.

D Additional results

Figure D1: Ex post security performance



Note: The panels of this figure show the kernel density of the raw (i.e. unadjusted) buy-and-hold abnormal returns (BHAR), split in security issues that were successful (solid line) versus issues that were not fully subscribed (dashed line). We consider every issuance above 90% subscription as successful. The BHARs are calculated over horizons of 0 (issuance month), 1, 6, and 12 months after issuance. Returns are computed from the simple average of high and low monthly prices from Tabellenboeken, benchmarked against the Dutch government bond on the run (for bonds) and a manually created stock price index from the selected set of stocks reported in the weekly Economisch Statistische Berichten (for stocks). All panels include only observations where the bank is one of the lead underwriters and the bank's guarantee does not exceed 5 million in the currency of issuance.

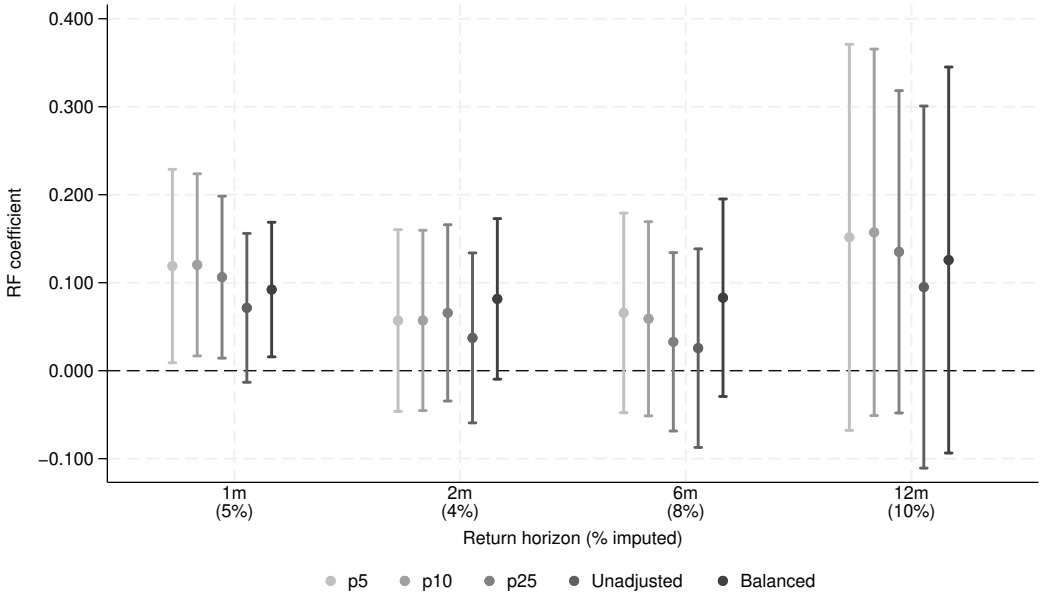
Table D1: Underwriting fees and banker guarantees

	(1) OLS	(2) RF	(3) IV (2nd stage)	(4) IV (1st stage)
	Underwriting fee (%)			Actual banker guarantee (x1000)
Actual banker guarantee (x1000)	−0.004		−0.004	
	(0.007)		(0.009)	
Predicted banker guarantee (x1000)		−0.003		0.837***
		(0.007)		(0.108)
Observations	1,328	1,328	1,328	1,328
R-squared	0.499	0.499	0.149	0.255
IV F-stat				60.43
Bin FE	Y	Y	Y	Y
Year FE	Y	Y	Y	Y
Industry FE	Y	Y	Y	Y
Director FE	Y	Y	Y	Y
Controls	Y	Y	Y	Y

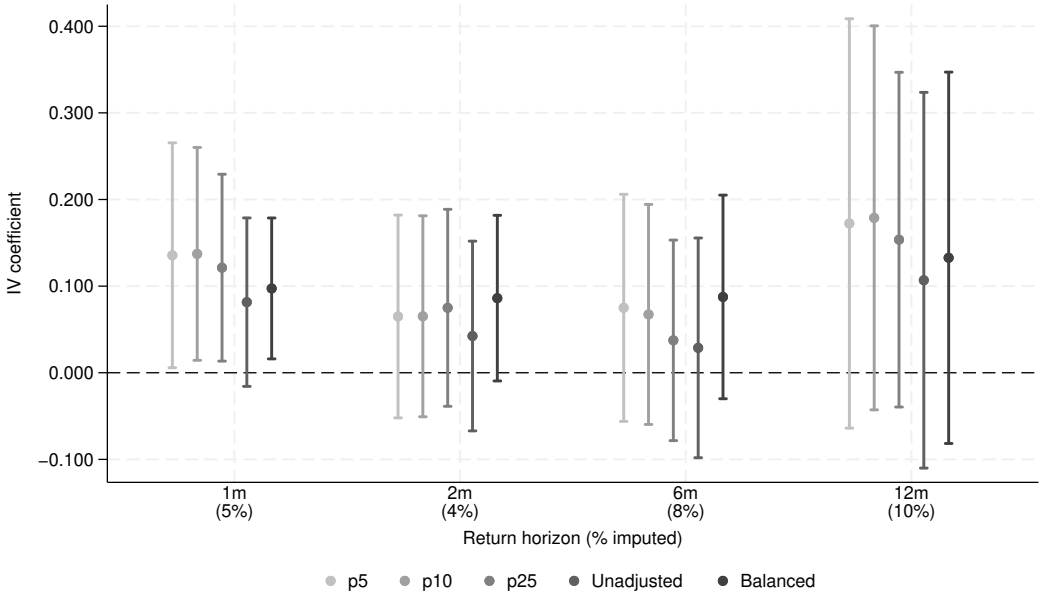
Note: This table presents estimates of actual and predicted banker underwriting for underwriting fees. We only include the observations for which the bank is the lead underwriter and where the total bank guarantee does not exceed 5 million in the currency of issuance. The underwriting fee is a percentage of the total guarantee of the bank. If the exact fee is not reported in the minutes, we take the spread between the buy and sell price of the securities agreed upon between the bank and the issuer. All specifications include the same set of controls (bank share in syndicate, a dummy for public issuers, stocks, foreign, interlocked directorate, a main bank dummy, an IPO dummy, and a rights-issue dummy). All columns have fixed effects: bin, year, industry and director. Standard errors (in parentheses) are clustered by issuance. The R-squared in the first-stage column is the within R-squared after absorbing the fixed effects. ***, ** and * indicate significance at 1, 5 and 10% level, respectively.

E Robustness

Figure E1: Coefficient plot for different imputations and BHAR horizons



(a) RF



(b) IV

Note:This figure displays the RF and IV coefficients of Equation 2 with 95% confidence interval for different buy-and-hold abnormal returns horizons. The BHARs are calculated over horizons of 0 (issuance month), 1, 6, and 12 months after issuance. The coefficients p5-p25 refer to missing values imputed with the fifth (25th) percentile of the distribution of either stocks or bonds in the 3 years around the issuance. Unadjusted includes all unadjusted BHARs. Balanced includes only securities that have a price for 12 consecutive months after issuance. Returns are computed from the simple average of high and low monthly prices from Tabellenboeken, and for 1919, 1920 and 1924 from the end of month daily security prices from the Prijscourant. All returns are benchmarked against the Dutch government bond on the run (for bonds) and a manually created stock price index from the selected set of stocks reported in the weekly Economisch Statistische Berichten (for stocks). All panels include only observations where the bank is one of the lead underwriters and the bank's guarantee does not exceed 5 million in the currency of issuance.

Table E1: Robustness: assignment of amounts on a bin midpoint

	(1) Midpoint to lower bin	(2) Midpoints excluded	(3) Midpoint to upper bin
		OLS	
Actual banker guarantee (x1000)	0.032***	0.030***	0.031***
	(0.008)	(0.008)	(0.008)
R^2	0.232	0.232	0.229
		RF	
Predicted banker guarantee (x1000)	0.020**	0.019*	0.018*
	(0.010)	(0.010)	(0.010)
R^2	0.198	0.201	0.196
		IV 2nd stage	
Actual banker guarantee (x1000)	0.025**	0.023*	0.022*
	(0.012)	(0.012)	(0.012)
R^2	0.069	0.068	0.066
		IV 1st stage Actual banker guarantee	
Predicted banker guarantee (x1000)	0.807***	0.801***	0.820***
	(0.107)	(0.107)	(0.107)
R^2	0.243	0.250	0.251
IV F-stat	56.76	55.89	58.83
Observations	1,391	1,321	1,391
Clusters	239	227	239
Mean of dep. var.	0.454	0.431	0.454
SD of Actual banker guarantee (x1000)	21.701	22.006	21.701
Bin FE	Y	Y	Y
Year FE	Y	Y	Y
Industry FE	Y	Y	Y
Director FE	Y	Y	Y
Controls	Y	Y	Y

Note: This table reports how the baseline estimates of Table 4, column (4), respond to the treatment of issuances whose bank guarantee falls exactly on a bin midpoint. Bins run from the midpoint of one step of the rule to the midpoint of the next, so an amount lying exactly halfway between two discontinuities is equidistant from both and its assignment is a convention rather than a feature of the data. Twelve issuances (70 director-observations) are affected: bank guarantees of 200, 300, 425 and 600 thousand in the 1919–1925 regime and 500 thousand in the 1925–1930 regime. Column (1) assigns them to the lower bin, as in the main specification. Column (2) excludes them. Column (3) assigns them to the upper bin instead. All twenty bins are populated under each of the three treatments, and the smallest non-singleton bin contains three issuances throughout. All specifications include the same set of controls (bank share in syndicate, a dummy for public issuers, stocks, foreign, interlocked directorate, a main bank dummy, an IPO dummy, and a rights-issue dummy) and have bin, year, industry and director fixed effects. Standard errors (in parentheses) are clustered by issuance. The R-squared reported in the first-stage panel is the within R-squared after absorbing the fixed effects, and the R-squared in the IV second stage is computed after partialling out the fixed effects; R-squared values are therefore not directly comparable across panels. ***, ** and * indicate significance at 1, 5 and 10% level, respectively.

F Counterfactual step functions

Sample and restrictions We treat the two periods separately. The sample consists of the issues above *f.* 150,000 without an interlocked director (32 in 1919–1925 and 146 in 1925–1930), with the median director guarantee.[40] In 1919–1925 we add the three issues above *f.* 2 million with an interlocked director in which every director personal guarantee was $f.100,000$. Above *f.* 2 million issues with an interlocked director follow the rule (see Section 4.2). The search imposes the restrictions of Section 2.2. Figure F1 shows the discontinuity of the banker guarantee around *f.* 150,000.

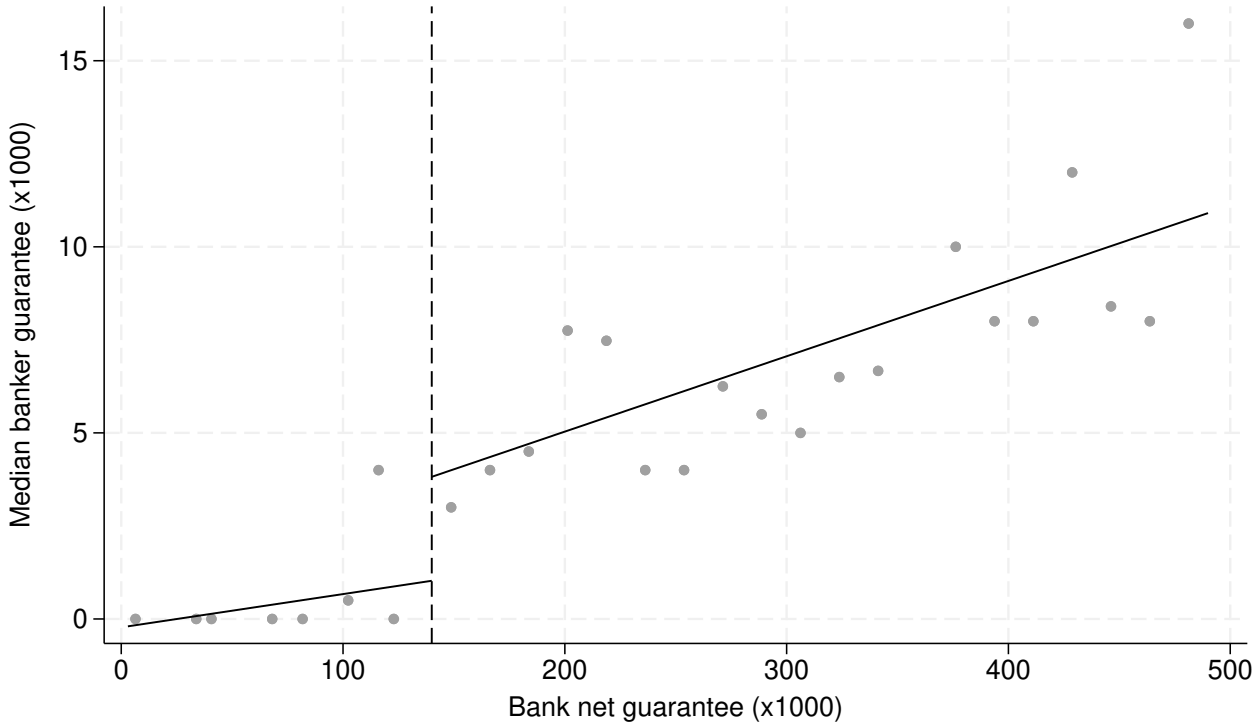
The rule originates in the period after October 1925, which contains 146 issues without an interlocked director and pins down the steps precisely. The period 1919–1925 contains only 32 such issues, too few to infer the step function from that period alone. We therefore take the personal guarantees that make up the step function from 1925–1930 and extend those steps to 1919–1925: a step is included when at least one issue without an interlocked director in 1919–1925 lies exactly on its level. The one exception is the top step (*f.* 100,000 for issues between *f.* 3.5 and 5 million), an personal guarantee that was not used after 1925. Above *f.* 2 million, issuances with an interlocked director do follow the rule: on four of the five such issues outside this step, every director guaranteed exactly the what the rule would predict. For this step we therefore also use the two issues with an interlocked director in which all four directors guaranteed *f.* 100,000. No issue in our sample exceeds *f.* 5 million, so the steps above that bank underwriting amount are an extrapolation and do not affect our estimates. Section 4.4 shows that an exhaustive search under these restrictions recovers our step function in both periods, and that data without a rule cannot reproduce its compliance. Figure 1 shows the resulting rule.

Dynamic Programme For a given dataset, we find the *provably best* step function among all step functions satisfying the restrictions. Candidate breakpoints lie on a grid of round *f.* 50,000 amounts. The restrictions are imposed during the search rather than checked afterwards. A dynamic programme over (breakpoint, width, level) states evaluates every admissible step function and returns the global optimum: the most issues exactly on the step function, with the smallest total absolute deviation as tie-breaker. The number of steps is chosen by the same criterion (up to 7/11 when capped, free otherwise). The optimum is exact over this candidate set rather than over all conceivable step functions. The grid is not binding for our purposes: every breakpoint of our own step function is a multiple of *f.* 50,000, so our rule lies inside the search space. Applied to the real data, the search returns almost exactly our rule (Figure F2).

[40] In two issues from 1919–1925 in which all directors personal guarantees were out of line with the president's (17 and 15 against 20 and 25), we set the director guarantee at 80% of the president's (16 and 20). One further guarantee in foreign currency is rounded to the nearest whole amount after conversion.

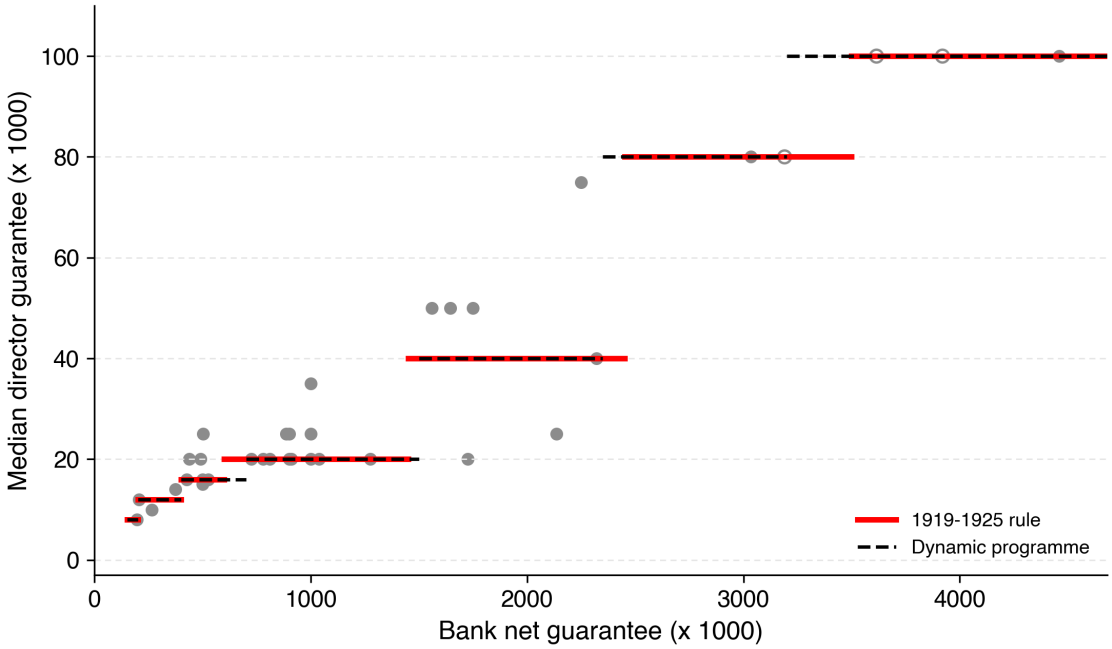
Counterfactual datasets Using our own data, we simulate 10,000 counterfactual banker guarantees per period. We estimate an OLS regression of banker guarantees on bank guarantees and store the intercept, slope and error terms. Since the errors are heteroskedastic, we impose the same error structure on the counterfactuals: we regress the squared residuals of the first regression on the bank guarantee, and draw counterfactual errors with mean zero and the fitted (observation-specific) variance. Counterfactual guarantees are the linear prediction plus the drawn error, rounded to the nearest actually observed banker guarantee; counterfactuals rounded to zero above f. 150,000 are excluded, as in the real data. Each counterfactual dataset is searched under the same restrictions as the real data. For 1925–1930 the candidate levels are the even personal guarantees observed at least twice in the counterfactual dataset itself, as in the real data; for 1919–1925 they are the same set as in the real data. The test statistic is the highest attainable share of issues exactly on a step function; the p-value is the share of counterfactual datasets that reach the compliance of our step function on the real data. The p-value is $(k+1)/(B+1)$, where k is the number of the $B = 10{,}000$ counterfactual datasets that reach the compliance of our step function on the real data.

Figure F1: Bank and bankers' guarantees discontinuity

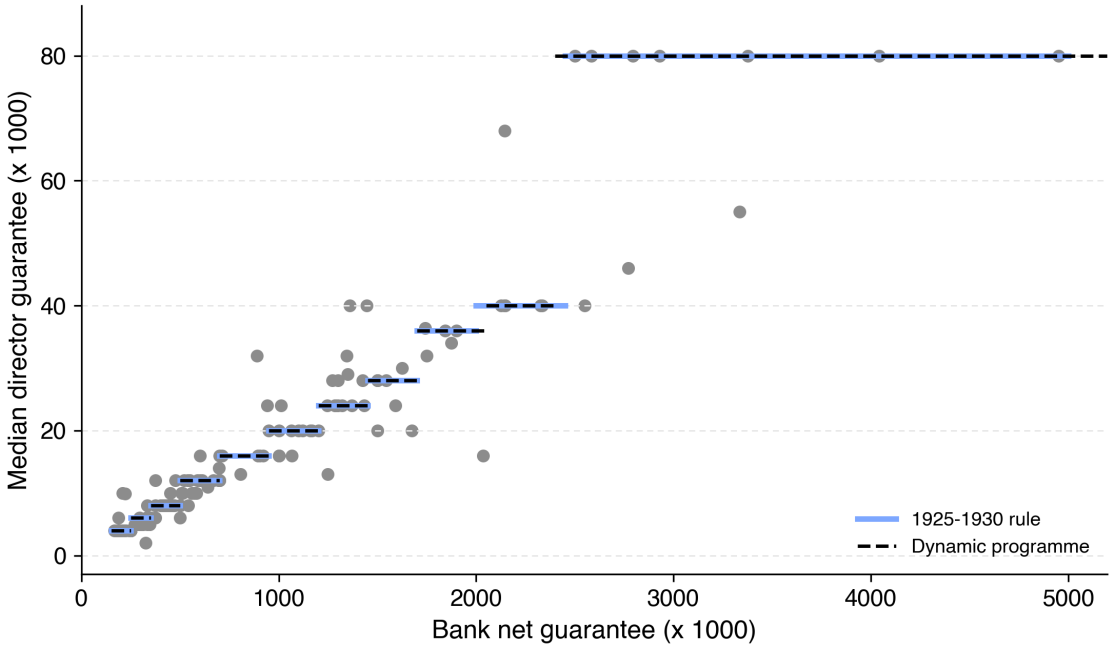


Note: This figure shows the RD plot of the bank net guarantee with a cutoff at f. 150,000. The RD plot has 8 bins below the cutoff and 20 above. Bank and banker guarantees are in thousands of currency of issuance. Includes only observations where the bank is one of the lead underwriters.

Figure F2: Dynamic programme vs. manual step function



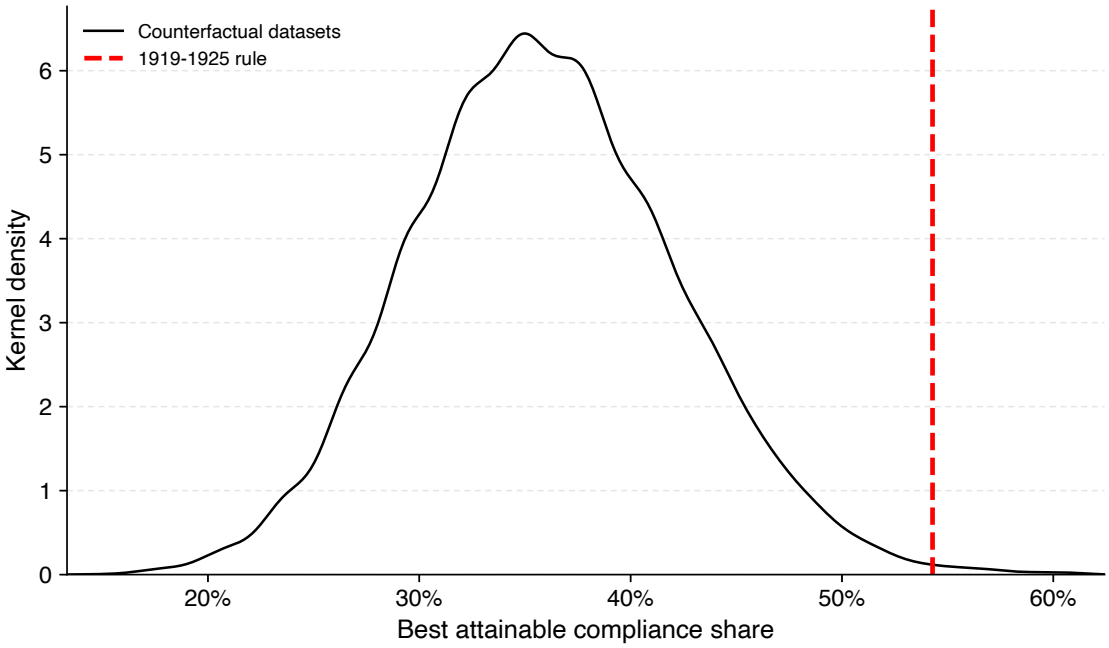
(a) Step function 1: 1919–1925



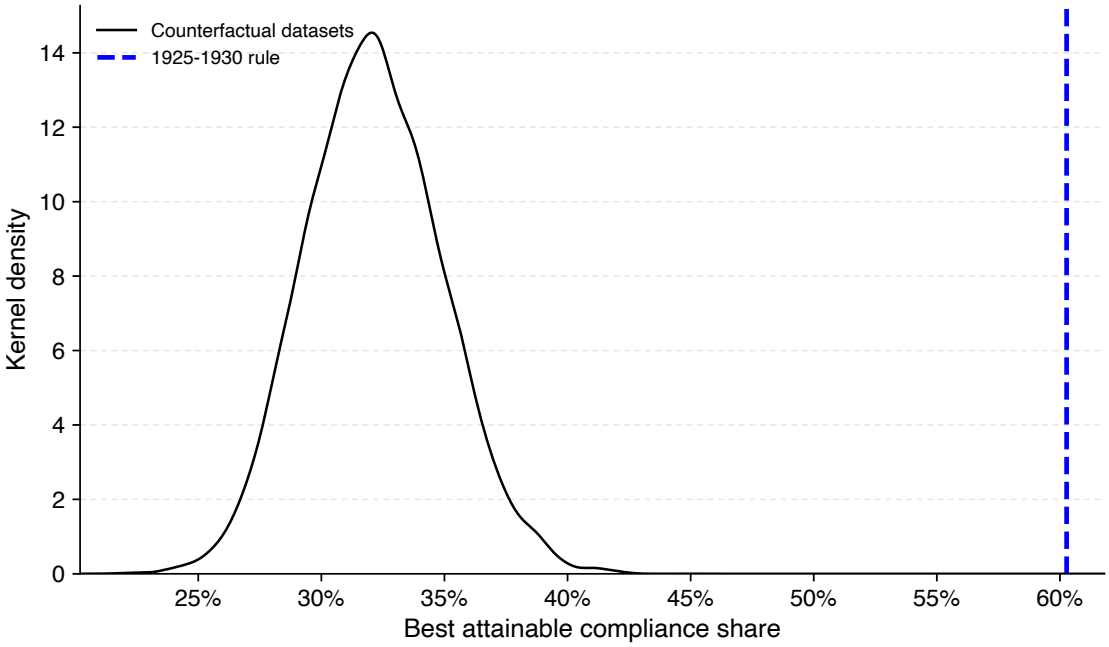
(b) Step function 2: 1925–1930

Note: Each circle is an issue underwritten by the bank, with the bank's net guarantee on the horizontal axis and the guarantee given by the bank's managers on the vertical axis, both in thousands; the 54 (Panel (a), 1919–1925) and 156 (Panel (b), 1925–1930) issues above the 150,000 floor are shown. The solid line is the guarantee rule inferred from the observed guarantees and used in the analysis. The dashed line is the step function that places the largest number of issues exactly on it among all step functions satisfying the restrictions in this appendix, found by dynamic programme and without using our inferred rule.

Figure F3: Counterfactual compliance



(a) Step function 1: 1919–1925



(b) Step function 2: 1925–1930

Note: Kernel density of the highest exact-compliance share attainable on each of 10,000 counterfactual datasets per period in which guarantees are set ad hoc, each fitted by the same dynamic programme as in Figure F2. The dashed line marks the compliance of the step function used in the analysis, 54% in Panel (a) and 60% in Panel (b).

Table F1: Admissible guarantee levels

	Steps (1919–25 / 1925–30)	Compliance (%)	First stage	KP F	RF	IV
Baseline rule (Table 4, col. 4)	7 / 11	48.6	0.807***	56.76	0.020**	0.025**
			(0.107)		(0.010)	(0.012)
Number of steps capped at 7 / 11						
All restrictions	7 / 11	48.6	0.780***	46.49	0.022**	0.028**
			(0.114)		(0.010)	(0.012)
Levels used twice within the period itself	5 / 11	47.7	0.668***	35.25	0.022**	0.033**
			(0.113)		(0.009)	(0.015)
Levels used twice over both periods	7 / 11	49.0	0.692***	38.52	0.025***	0.036**
			(0.111)		(0.010)	(0.015)
Odd levels allowed	7 / 11	48.5	0.779***	46.58	0.022**	0.028**
			(0.114)		(0.010)	(0.012)
Number of steps chosen by the search						
All restrictions	7 / 11	48.6	0.780***	46.49	0.022**	0.028**
			(0.114)		(0.010)	(0.012)
Levels used twice within the period itself	5 / 11	47.7	0.668***	35.25	0.022**	0.033**
			(0.113)		(0.009)	(0.015)
Levels used twice over both periods	8 / 11	49.4	0.690***	37.97	0.025***	0.037**
			(0.112)		(0.010)	(0.015)
Odd levels allowed	7 / 11	48.5	0.779***	46.58	0.022**	0.028**
			(0.114)		(0.010)	(0.012)
Observations: 1,391; clusters: 239						

Note: This table changes only the admissible levels, the guarantee amounts the search may use, keeping all other restrictions. The first row is the baseline rule, used throughout the paper. The second applies all restrictions of Section 2.2 and lets an exhaustive search over all admissible step functions choose the thresholds and levels (Section 4.4, Appendix F): the rule is inferred from issues without an interlocked director, plus the three 1919–1925 issues above *f.* 2 million in which every director guaranteed the same amount; steps are anchored (at least one issue exactly on each step's level); admissible levels are even amounts used at least twice in 1925–1930. Under these restrictions the search returns the baseline rule: it predicts the same guarantee for all issues but one, and differs only in thresholds that fall in ranges without issues. In the third row each period's levels are the even amounts used at least twice within that period; for 1919–1925 this leaves only five levels, as few issues remain. In the fourth row levels are the even amounts used at least twice over both periods together. In the fifth row levels are all amounts, odd ones included, used at least twice in 1925–1930. The upper panel caps the number of steps at that of the baseline rule, the lower panel lets the search choose it. All rows re-estimate Table 4, column (4), with its controls and fixed effects, replacing only the rule that defines the predicted banker guarantee; the president and secretary are scaled at 125 and 75% of the median director as in the baseline rule unless stated. Each rule comes with its own bin fixed effects, built from midpoint to midpoint around its own thresholds as those of the baseline rule are. Steps gives the number of steps between the floor and *f.* 5 million, the range of our sample, in 1919–1925 and 1925–1930; the baseline rule adds one extrapolated step above *f.* 5 million in each period, which no issue in the sample reaches. Compliance is the share of banker-observations in the regression sample, interlocked issues included, whose guarantee equals the rule's prediction; it therefore differs from the issue-level shares in Section 4.4, which use the median director on issues without an interlocked director. Standard errors (in parentheses) are clustered by issuance. ***, ** and * indicate significance at 1, 5 and 10% level, respectively.

Table F2: Relaxing one restriction at a time

	Steps (1919–25 / 1925–30)	Compliance (%)	First stage	KP F	RF	IV
Baseline rule (Table 4, col. 4)	7 / 11	48.6	0.807***	56.76	0.020**	0.025**
			(0.107)		(0.010)	(0.012)
Number of steps capped at 7 / 11						
All restrictions	7 / 11	48.6	0.780***	46.49	0.022**	0.028**
			(0.114)		(0.010)	(0.012)
Anchored steps dropped	7 / 11	48.6	0.780***	46.49	0.022**	0.028**
			(0.114)		(0.010)	(0.012)
Floor chosen by the search	7 / 11	48.6	0.780***	46.49	0.022**	0.028**
			(0.114)		(0.010)	(0.012)
Widening steps dropped	7 / 11	49.0	0.864***	41.35	0.005	0.006
			(0.134)		(0.009)	(0.010)
Interlocked issues included	7 / 11	48.5	0.830***	54.83	0.001	0.001
			(0.112)		(0.009)	(0.011)
Salary ratios ignored	7 / 11	44.3	0.744***	34.00	0.020*	0.027*
			(0.128)		(0.011)	(0.015)
Monotonicity dropped	7 / 11	48.5	0.550***	11.81	0.013	0.023
			(0.160)		(0.010)	(0.016)
Number of steps chosen by the search						
All restrictions	7 / 11	48.6	0.780***	46.49	0.022**	0.028**
			(0.114)		(0.010)	(0.012)
Anchored steps dropped	7 / 11	48.6	0.780***	46.49	0.022**	0.028**
			(0.114)		(0.010)	(0.012)
Floor chosen by the search	7 / 11	48.6	0.780***	46.49	0.022**	0.028**
			(0.114)		(0.010)	(0.012)
Widening steps dropped	7 / 12	49.4	0.865***	40.34	0.006	0.006
			(0.136)		(0.009)	(0.010)
Interlocked issues included	7 / 11	48.5	0.830***	54.83	0.001	0.001
			(0.112)		(0.009)	(0.011)
Salary ratios ignored	7 / 11	44.3	0.744***	34.00	0.020*	0.027*
			(0.128)		(0.011)	(0.015)
Monotonicity dropped	10 / 25	50.8	0.920***	49.80	0.011	0.012
			(0.130)		(0.010)	(0.011)
Observations: 1,391; clusters: 239						

Note: This table drops the restrictions behind the rule one at a time, keeping all the others, so that each row shows the effect of that restriction alone. The first row is the baseline rule, used throughout the paper. The second applies all restrictions of Section 2.2 and lets an exhaustive search over all admissible step functions choose the thresholds and levels (Section 4.4, Appendix F): the rule is inferred from issues without an interlocked director, plus the three 1919–1925 issues above *f.* 2 million in which every director guaranteed the same amount; steps are anchored (at least one issue exactly on each step's level); admissible levels are even amounts used at least twice in 1925–1930. Under these restrictions the search returns the baseline rule: it predicts the same guarantee for all issues but one, and differs only in thresholds that fall in ranges without issues. Each following row drops a single restriction: anchored steps (no issue needs to lie on a step's level); the floor (the search chooses it instead of *f.* 150,000); widening steps (steps may narrow); no interlocks (the search is fitted to Bierens' own guarantee on all issues, including those with an interlocked director); fixed salary ratios (instead of setting the president's and secretary's guarantees at 125% and 75% of a director's, the search fits a separate step function for the president, the directors and the secretary, for which Steps refers to the directors' schedule); and monotonicity (guarantee levels may fall from one step to the next). No overlap is always kept. Estimating the salary ratios from the data instead of imposing them is not shown: the most frequent ratios in the data are exactly 125, 75 and 50% (junior director), so the rule is unchanged. The upper panel caps the number of steps at that of the baseline rule, the lower panel lets the search choose it. All rows re-estimate Table 4, column (4), with its controls and fixed effects, replacing only the rule that defines the predicted banker guarantee; the president and secretary are scaled at 125 and 75% of the median director as in the baseline rule unless stated. Each rule comes with its own bin fixed effects, built from midpoint to midpoint around its own thresholds as those of the baseline rule are. Steps gives the number of steps between the floor and *f.* 5 million, the range of our sample, in 1919–1925 and 1925–1930; the baseline rule adds one extrapolated step above *f.* 5 million in each period, which no issue in the sample reaches. Compliance is the share of banker-observations in the regression sample, interlocked issues included, whose guarantee equals the rule's prediction; it therefore differs from the issue-level shares in Section 4.4, which use the median director on issues without an interlocked director. Standard errors (in parentheses) are clustered by issuance. ***, ** and * indicate significance at 1, 5 and 10% level, respectively.

Table F3: Relaxing the restrictions behind the rule cumulatively

	Steps (1919–25 / 1925–30)	Compliance (%)	First stage	KP F	RF	IV
Baseline rule (Table 4, col. 4)	7 / 11	48.6	0.807***	56.76	0.020**	0.025**
			(0.107)		(0.010)	(0.012)
Number of steps capped at 7 / 11						
All restrictions	7 / 11	48.6	0.780***	46.49	0.022**	0.028**
			(0.114)		(0.010)	(0.012)
Anchored steps dropped	7 / 11	48.6	0.780***	46.49	0.022**	0.028**
			(0.114)		(0.010)	(0.012)
+ floor chosen by the search	7 / 11	48.6	0.780***	46.49	0.022**	0.028**
			(0.114)		(0.010)	(0.012)
+ widening steps dropped	7 / 11	49.0	0.864***	40.62	0.006	0.008
			(0.136)		(0.009)	(0.010)
+ interlocked issues included	7 / 11	49.0	0.761***	46.13	−0.008	−0.010
			(0.112)		(0.009)	(0.013)
+ salary ratios ignored	7 / 11	45.4	0.669***	18.47	0.003	0.005
			(0.156)		(0.012)	(0.018)
+ monotonicity dropped	7 / 11	45.9	0.611***	24.63	−0.005	−0.008
			(0.123)		(0.009)	(0.015)
Number of steps chosen by the search						
All restrictions	7 / 11	48.6	0.780***	46.49	0.022**	0.028**
			(0.114)		(0.010)	(0.012)
Anchored steps dropped	7 / 11	48.6	0.780***	46.49	0.022**	0.028**
			(0.114)		(0.010)	(0.012)
+ floor chosen by the search	7 / 11	48.6	0.780***	46.49	0.022**	0.028**
			(0.114)		(0.010)	(0.012)
+ widening steps dropped	8 / 12	49.3	0.893***	52.60	0.003	0.004
			(0.123)		(0.008)	(0.009)
+ interlocked issues included	7 / 12	49.4	0.761***	45.45	−0.007	−0.010
			(0.113)		(0.009)	(0.013)
+ salary ratios ignored	7 / 12	45.7	0.668***	18.15	0.004	0.005
			(0.157)		(0.012)	(0.018)
+ monotonicity dropped	13 / 31	48.4	0.603***	40.99	−0.005	−0.009
			(0.094)		(0.010)	(0.017)
Observations: 1,377–1,391; clusters: 239						

Note: This table relaxes the restrictions behind the rule cumulatively: each row drops one more restriction on top of those dropped above it. The first row is the baseline rule, used throughout the paper. The second applies all restrictions of Section 2.2 and lets an exhaustive search over all admissible step functions choose the thresholds and levels (Section 4.4, Appendix F): the rule is inferred from issues without an interlocked director, plus the three 1919–1925 issues above *f.* 2 million in which every director guaranteed the same amount; steps are anchored (at least one issue exactly on each step's level); admissible levels are even amounts used at least twice in 1925–1930. Under these restrictions the search returns the baseline rule: it predicts the same guarantee for all issues but one, and differs only in thresholds that fall in ranges without issues. The restrictions are then dropped in this order: anchored steps (no issue needs to lie on a step's level); the floor (the search chooses it instead of *f.* 150,000); widening steps (steps may narrow); no interlocks (the search is fitted to Bierens' own guarantee on all issues, including those with an interlocked director); fixed salary ratios (instead of setting the president's and secretary's guarantees at 125% and 75% of a director's, the search fits a separate step function for the president, the directors and the secretary, for which Steps refers to the directors' schedule); and monotonicity (guarantee levels may fall from one step to the next). No overlap is always kept. Estimating the salary ratios from the data instead of imposing them is not shown: the most frequent ratios in the data are exactly 125, 75 and 50% (junior director), so the rule is unchanged. In the last row the president, the directors and the secretary each have their own bins, and some bins contain a single observation; these singletons are dropped, which reduces the sample to 1,389 (capped) and 1,377 (free) observations. The upper panel caps the number of steps at that of the baseline rule, the lower panel lets the search choose it. All rows re-estimate Table 4, column (4), with its controls and fixed effects, replacing only the rule that defines the predicted banker guarantee; the president and secretary are scaled at 125 and 75% of the median director as in the baseline rule unless stated. Each rule comes with its own bin fixed effects, built from midpoint to midpoint around its own thresholds as those of the baseline rule are. Steps gives the number of steps between the floor and *f.* 5 million, the range of our sample, in 1919–1925 and 1925–1930; the baseline rule adds one extrapolated step above *f.* 5 million in each period, which no issue in the sample reaches. Compliance is the share of banker-observations in the regression sample, interlocked issues included, whose guarantee equals the rule's prediction; it therefore differs from the issue-level shares in Section 4.4, which use the median director on issues without an interlocked director. Standard errors (in parentheses) are clustered by issuance. ***, ** and * indicate significance at 1, 5 and 10% level, respectively.